Electronic Devices and Circuit Theory

Today's Topics:

- Semiconductors
- Diodes

Important!

- **Quiz 1 will be next week!**

The date is 26th March – at 5pm

Classrooms: Z11 and 103

- **The lecture on 23rd March will be held between 4-6pm.**

Non-linear Circuits

Mainly comprised of:

- **Diodes**
- **Transistors**

They are made with semiconductor materials.

Semiconductors

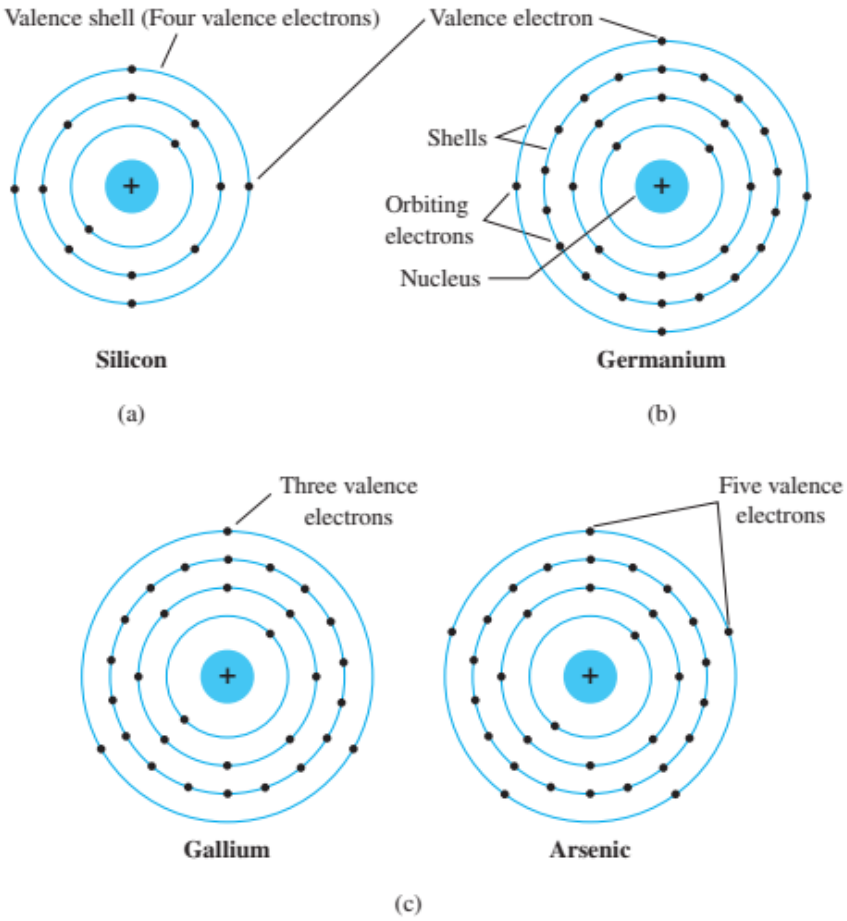


FIG. 1.3
Atomic structure of (a) silicon; (b) germanium; and (c) gallium and arsenic.

Semiconductors

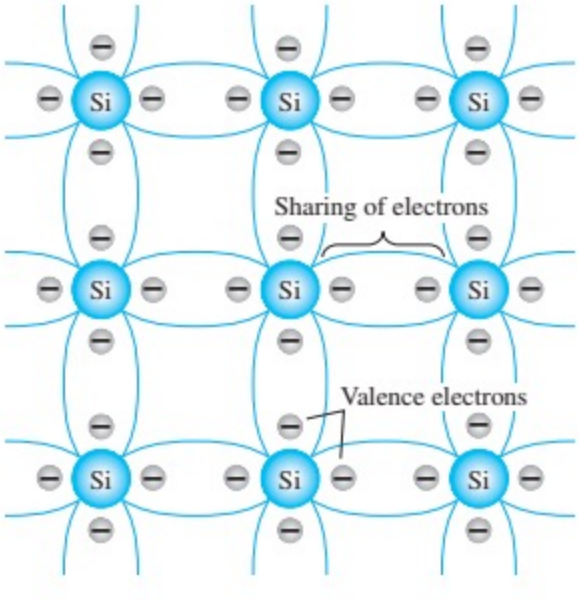


FIG. 1.4

Covalent bonding of the silicon atom.

Semiconductors

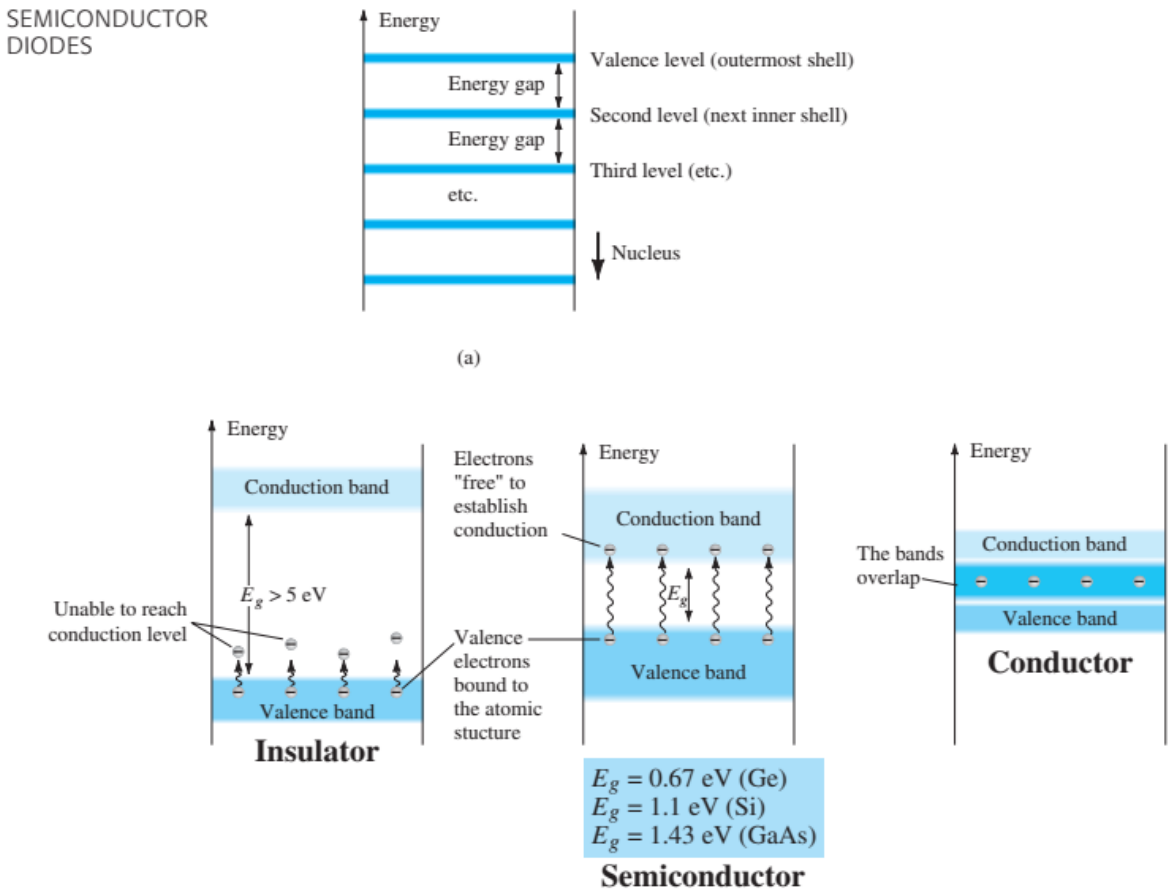


FIG. 1.6

Energy levels: (a) discrete levels in isolated atomic structures; (b) conduction and valence bands of an insulator, a semiconductor, and a conductor.

n-type material

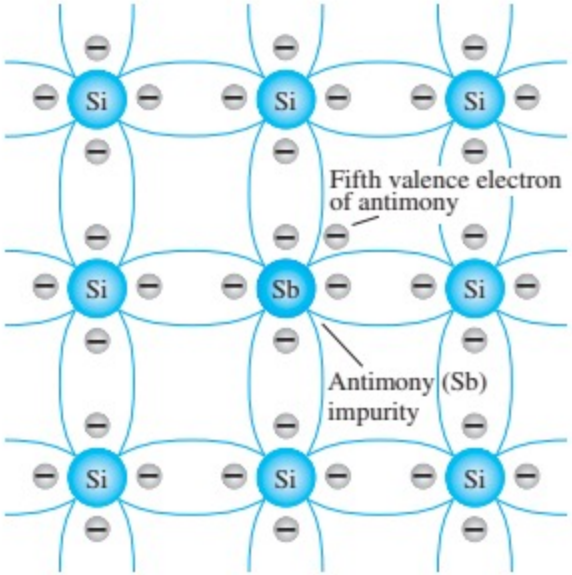


FIG. 1.7

Antimony impurity in n-type material.

n-type material

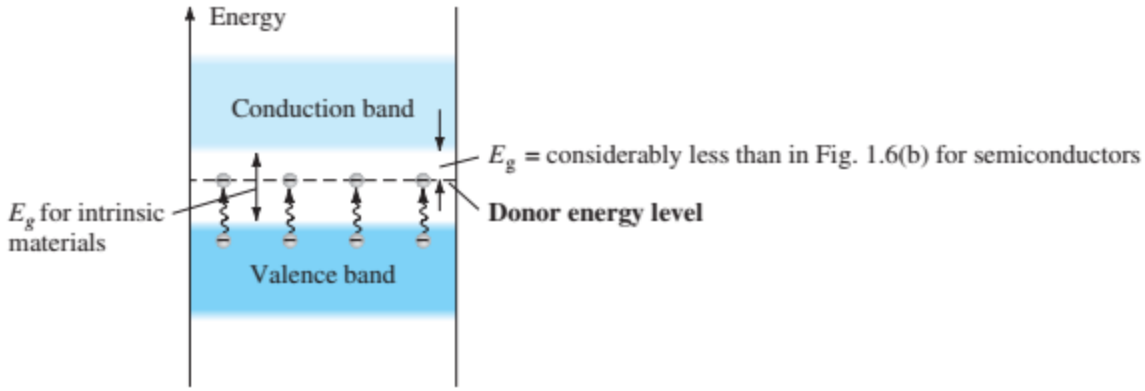


FIG. 1.8
Effect of donor impurities on the energy band structure.

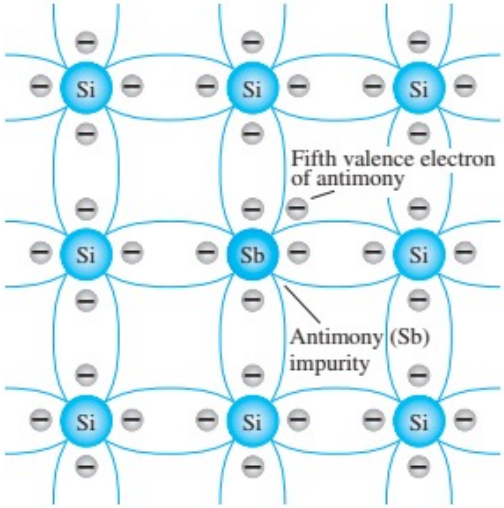


FIG. 1.7
Antimony impurity in n-type material.

p-type material

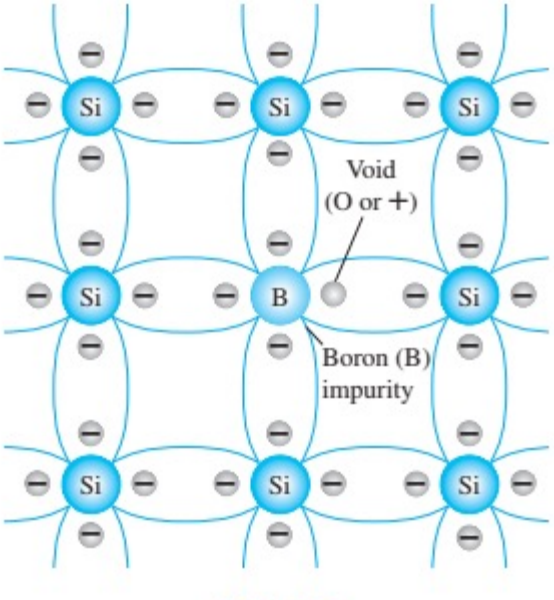


FIG. 1.9
Boron impurity in p-type material.

p-type material

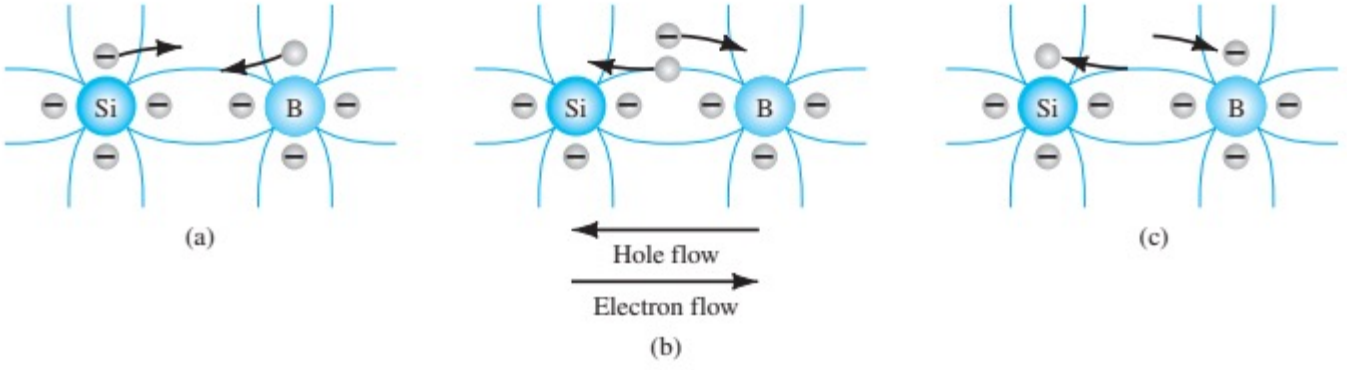
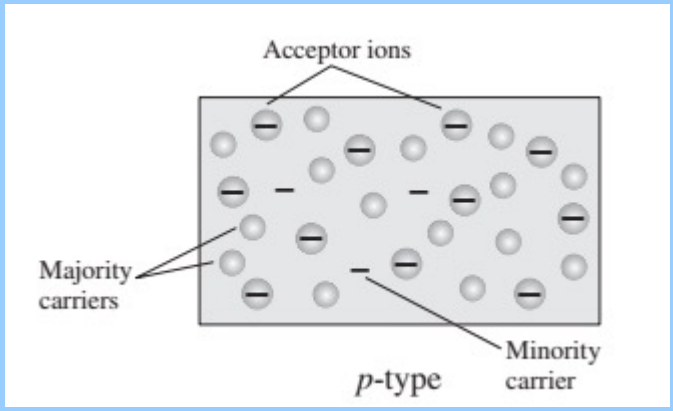
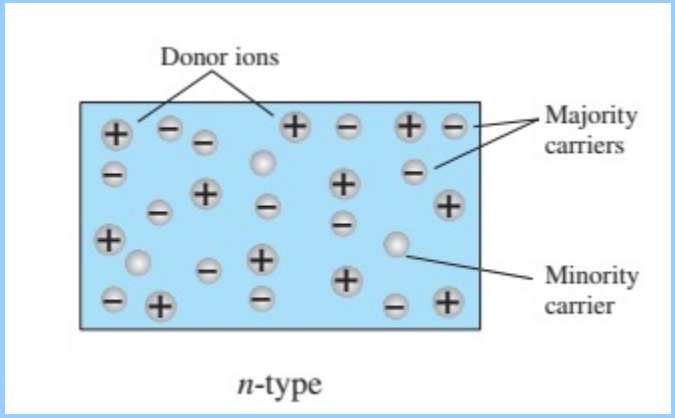


FIG. 1.10
Electron versus hole flow.

Semiconductors





- ***p*-type materials contain an excess of valence band holes.**
- ***n*-type materials contain an excess of conduction band electrons.**

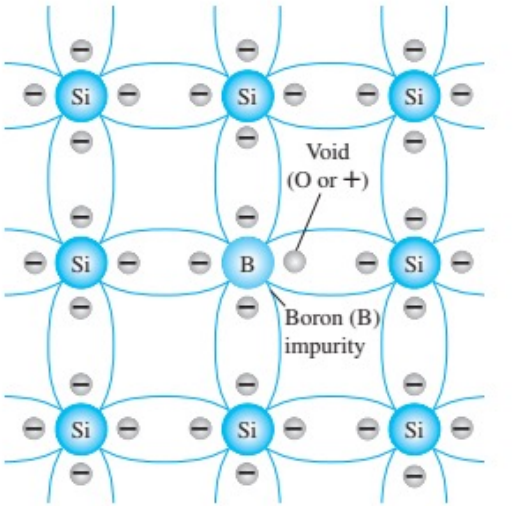


FIG. 1.9
Boron impurity in p-type material.

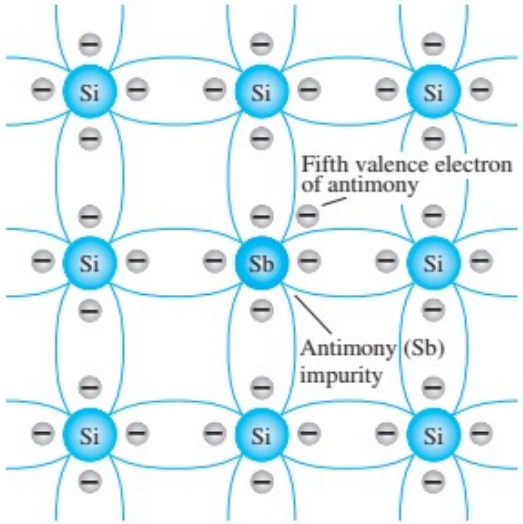
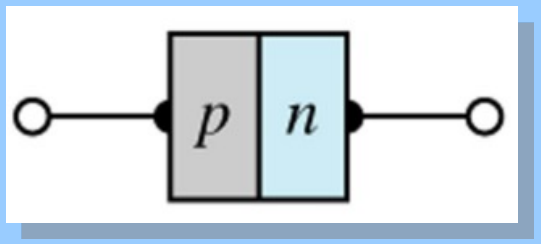


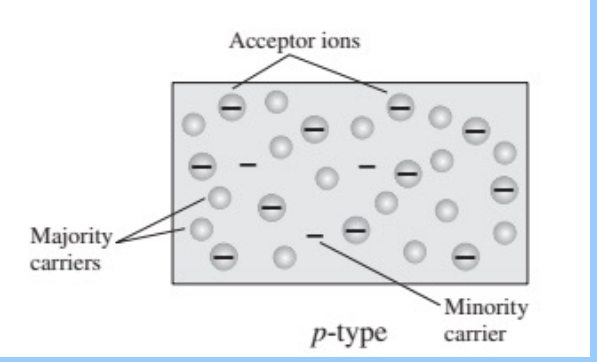
FIG. 1.7
Antimony impurity in n-type material.

p-n Junctions

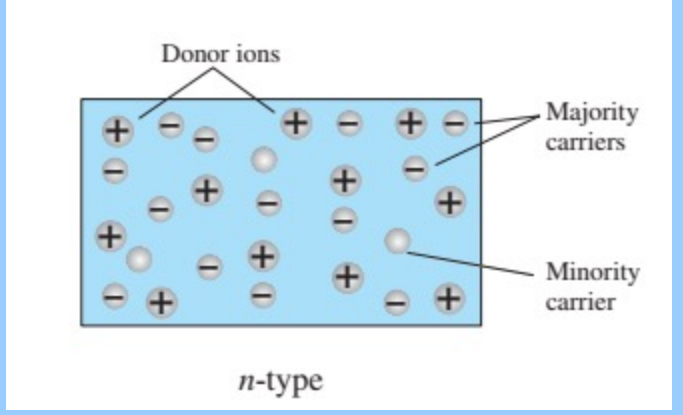
One end of a silicon or germanium crystal can be doped as a *p*-type material and the other end as an *n*-type material.

The result is a *p-n* junction.

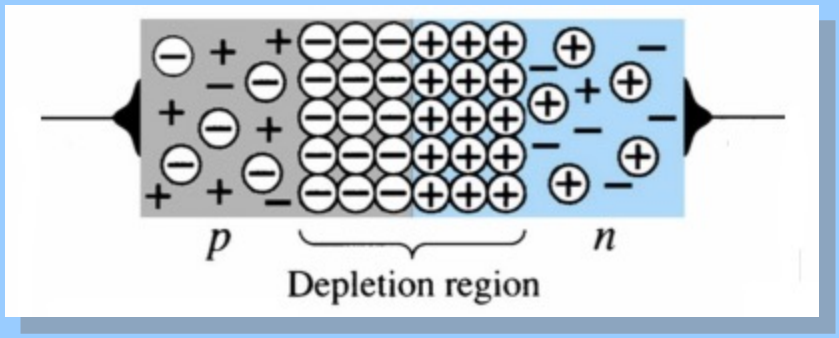




p-n Junctions



At the *p-n* junction, the excess conduction-band electrons on the *n*-type side are attracted to the valence-band holes on the *p*-type side.



The electrons in the *n*-type material migrate across the junction to the *p*-type material (electron flow).

The result is the formation of a depletion region around the junction.

The electron migration results in a negative charge on the p-type side of the junction and a positive charge on the n-type side of the junction.

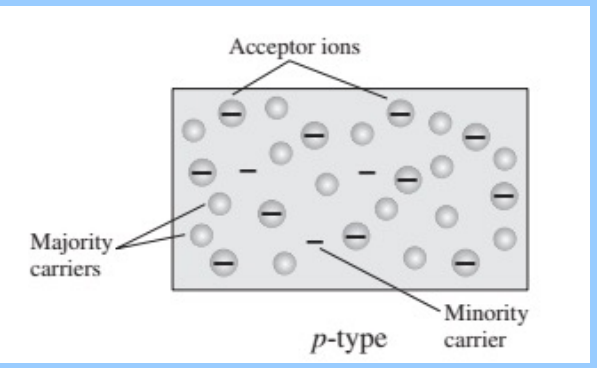
Diode Operating Conditions

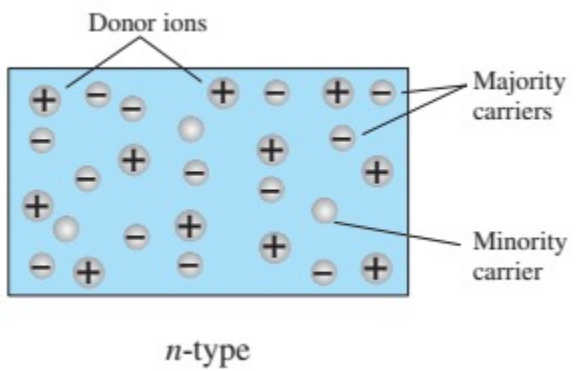
A diode has three operating conditions:

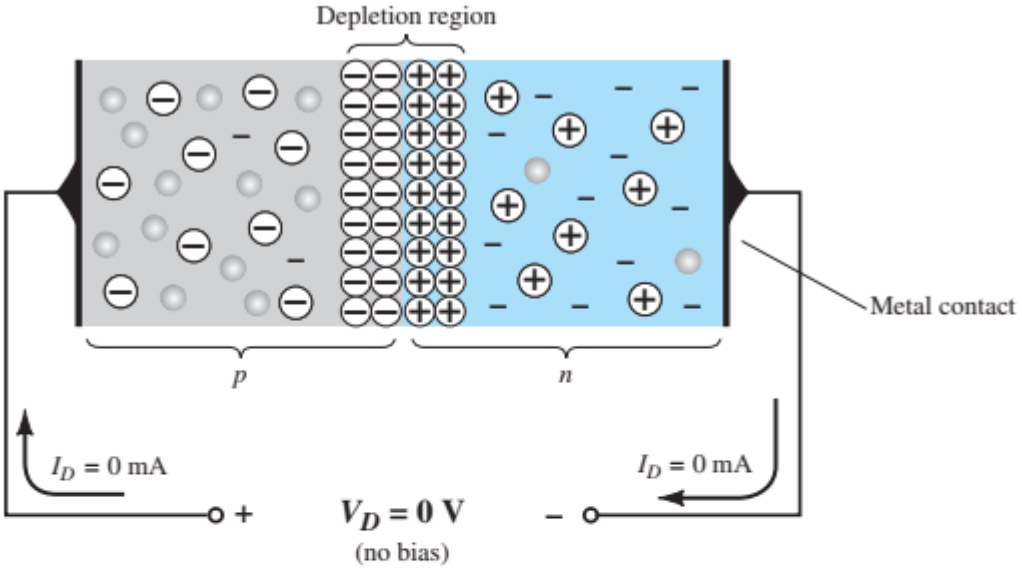
- **No bias**
- **Forward bias**
- **Reverse bias**

Diode

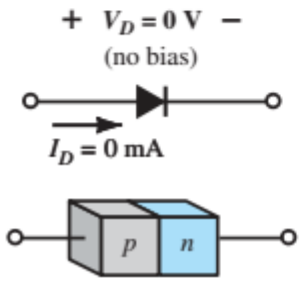
$V_D = 0V$

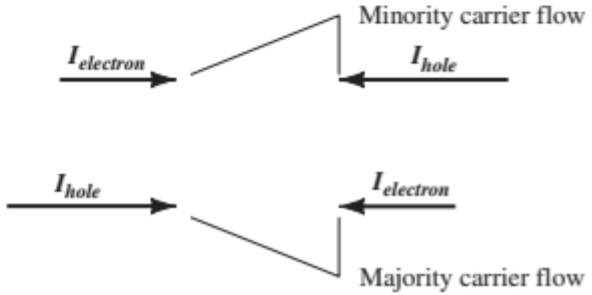

Acceptor ions
Majority carriers
Minority carrier
p-type


Donor ions
Majority carriers
Minority carrier
n-type


Depletion region
Metal contact
p
n
I_D = 0 mA
I_D = 0 mA
+
V_D = 0 V
(no bias)
–

(a)

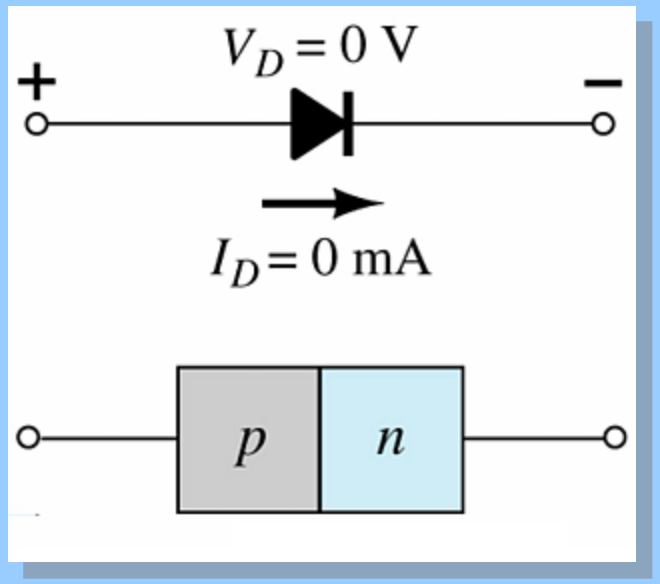

+ V_D = 0 V –
(no bias)
I_D = 0 mA
p
n


Minority carrier flow
$I_{electron}$
I_{hole}
I_{hole}
$I_{electron}$
Majority carrier flow

Diode Operating Conditions

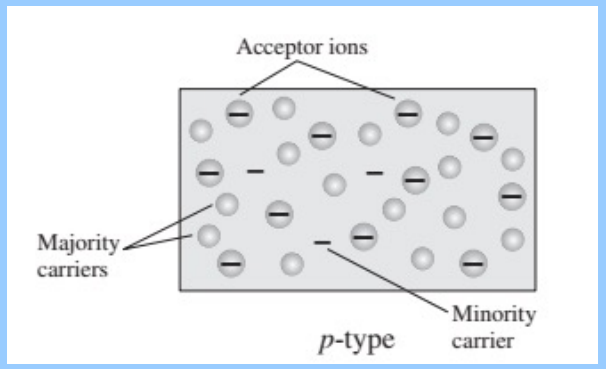
No Bias (V_D= 0V)

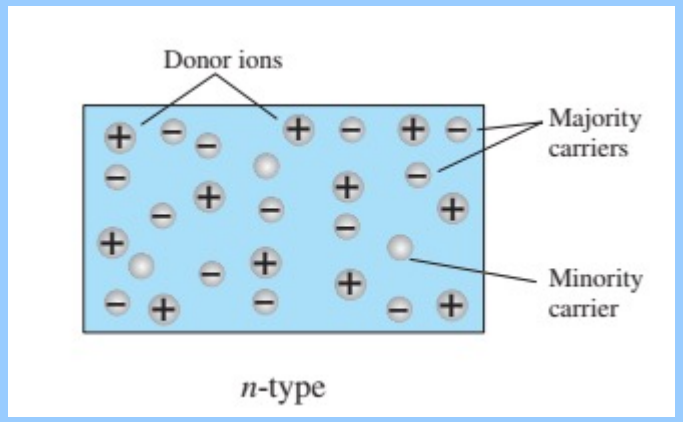
- **No external voltage is applied: V_D = 0 V**
- **No current is flowing: I_D = 0 A**
- **Only a modest depletion region exists**

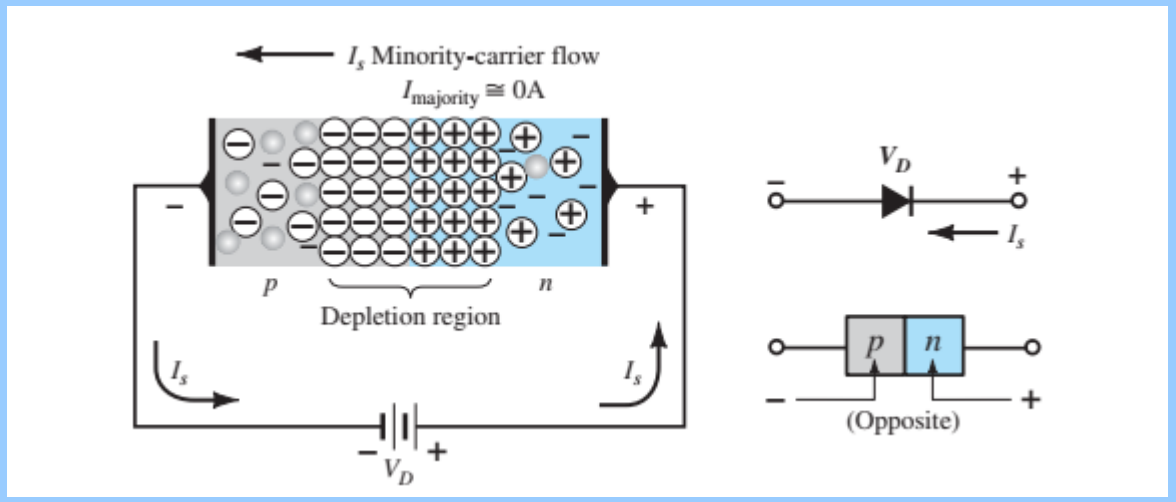


Diode

$V_D < 0V$


Acceptor ions
Majority carriers
Minority carrier
p-type

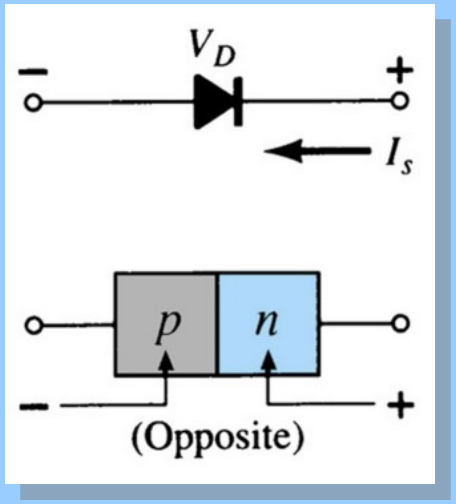

Donor ions
Majority carriers
Minority carrier
n-type

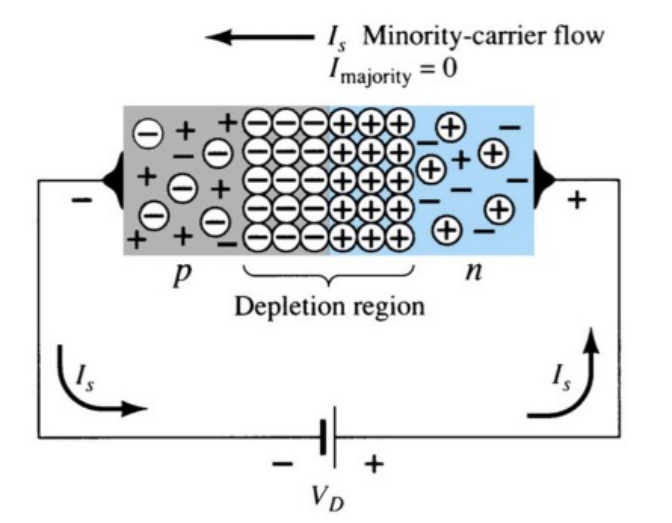

I_s Minority-carrier flow
$I_{majority} \cong 0A$
p
n
Depletion region
I_s
I_s
V_D
V_D
I_s
p
n
(Opposite)

Diode Operating Conditions

Reverse Bias ($V_D < 0V$)

External voltage is applied across the *p*-*n* junction in the opposite polarity of the *p*- and *n*-type materials.

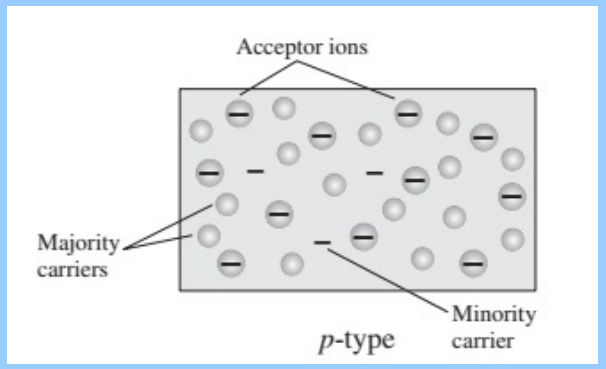


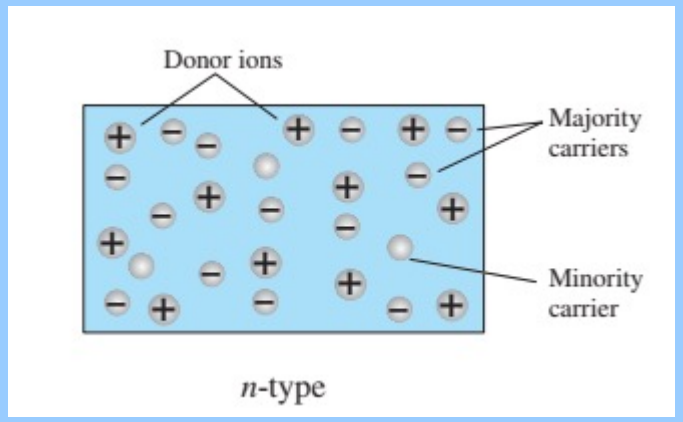


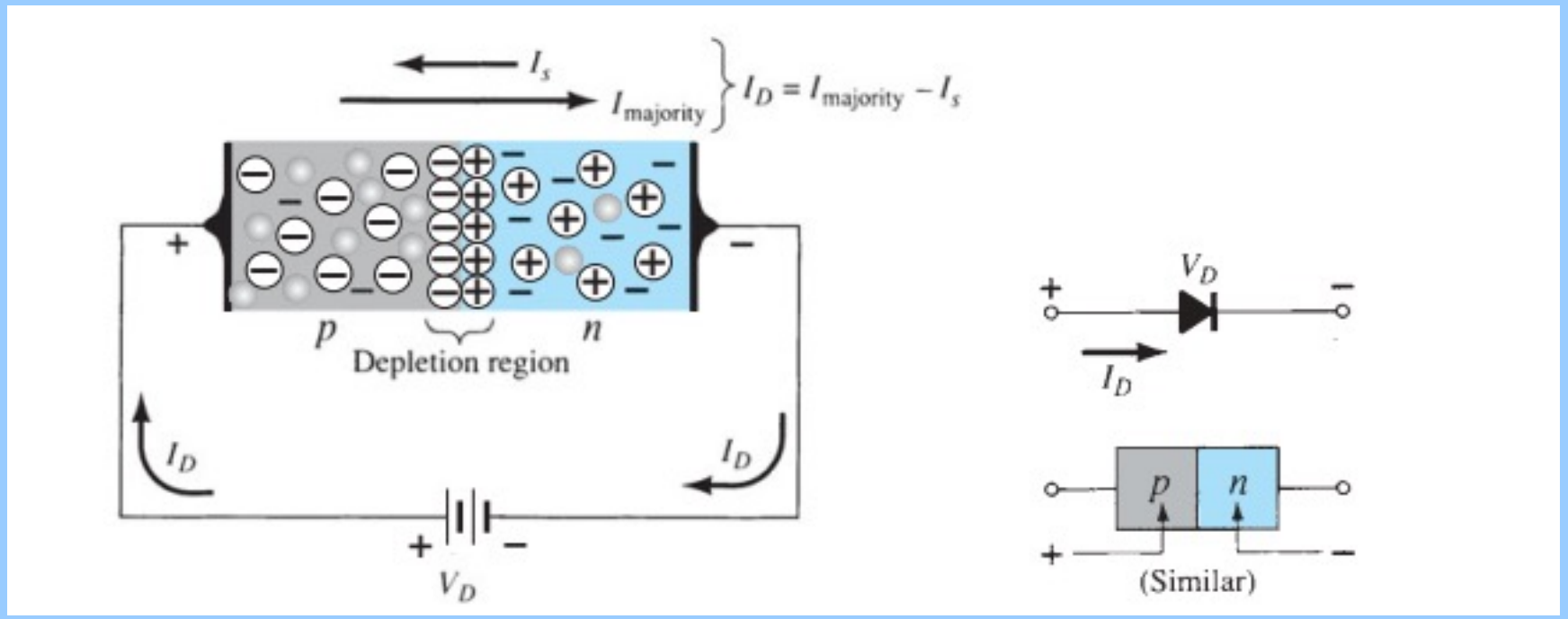
- **The reverse voltage causes the depletion region to widen.**
- **The electrons in the *n*-type material are attracted toward the positive terminal of the voltage source.**
- **The holes in the *p*-type material are attracted toward the negative terminal of the voltage source.**

Diode

$V_D > 0V$


Acceptor ions
Majority carriers
Minority carrier
p-type

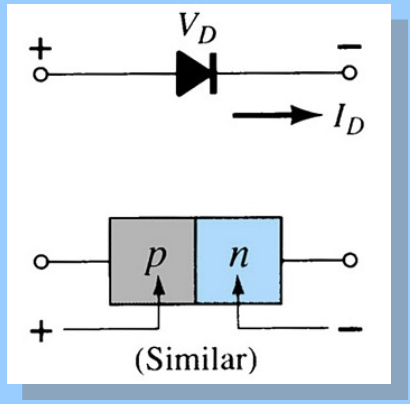

Donor ions
Majority carriers
Minority carrier
n-type

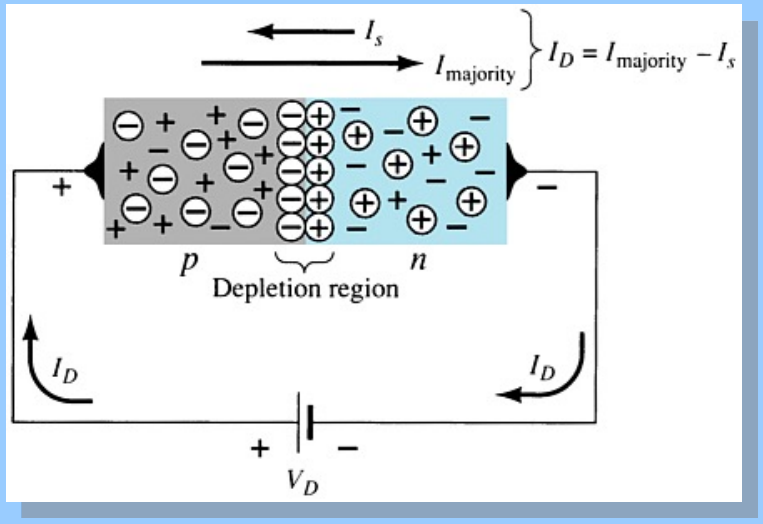

I_s
$I_{majority}$
$I_D = I_{majority} - I_s$
p
n
Depletion region
I_D
I_D
V_D
V_D
I_D
p
n
(Similar)

Diode Operating Conditions

Forward Bias ($V_D < 0V$)

External voltage is applied across the *p-n* junction in the same polarity as the *p*- and *n*-type materials.



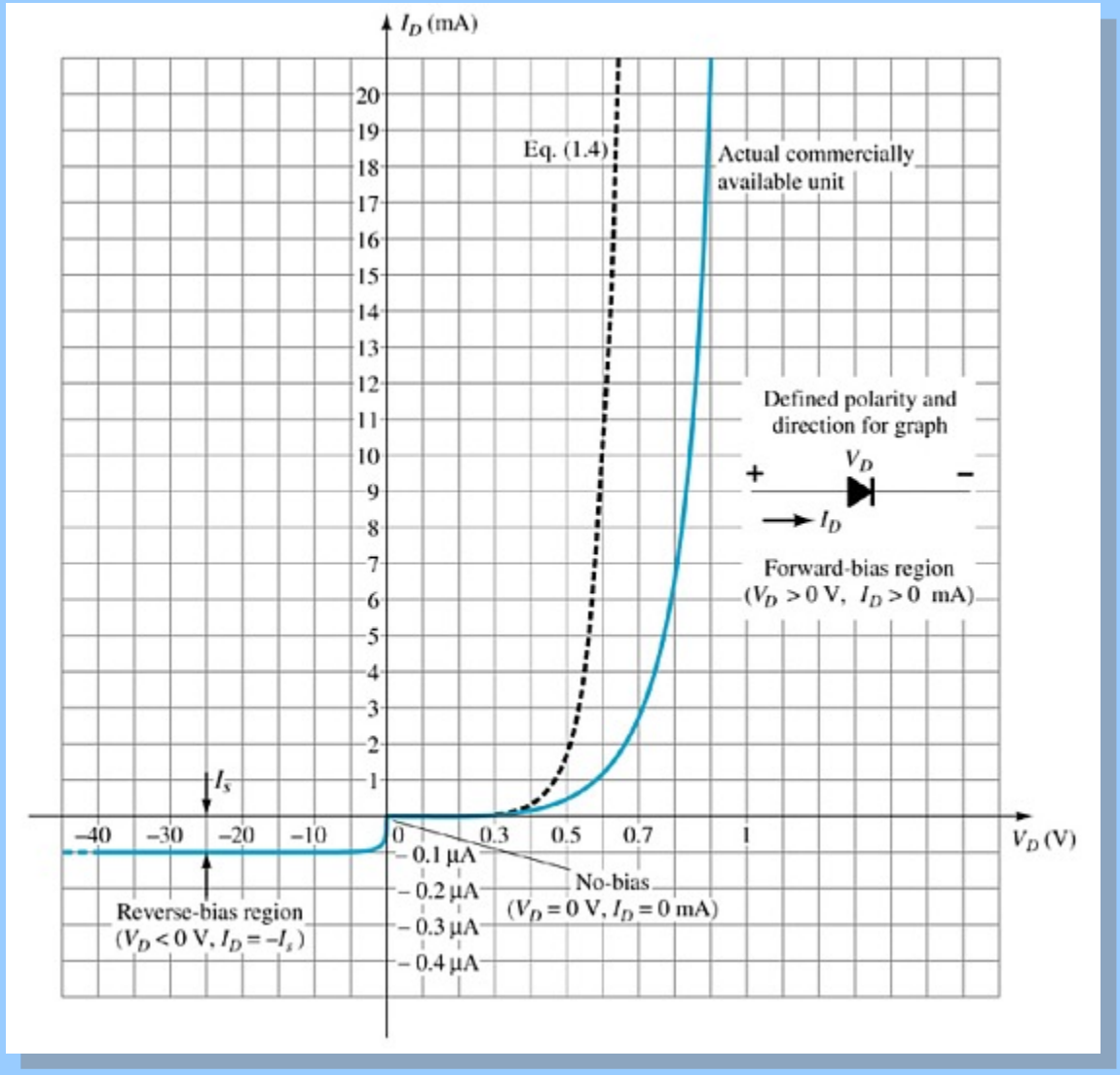


- **The forward voltage causes the depletion region to narrow.**
- **The electrons and holes are pushed toward the *p-n* junction.**

Actual Diode Characteristics

Note the regions for no bias, reverse bias, and forward bias conditions.

Carefully note the scale for each of these conditions.



Actual Diode Characteristics

$$I_D = I_s(e^{V_D/nV_T} - 1) \quad \text{(A)} \tag{1.2}$$

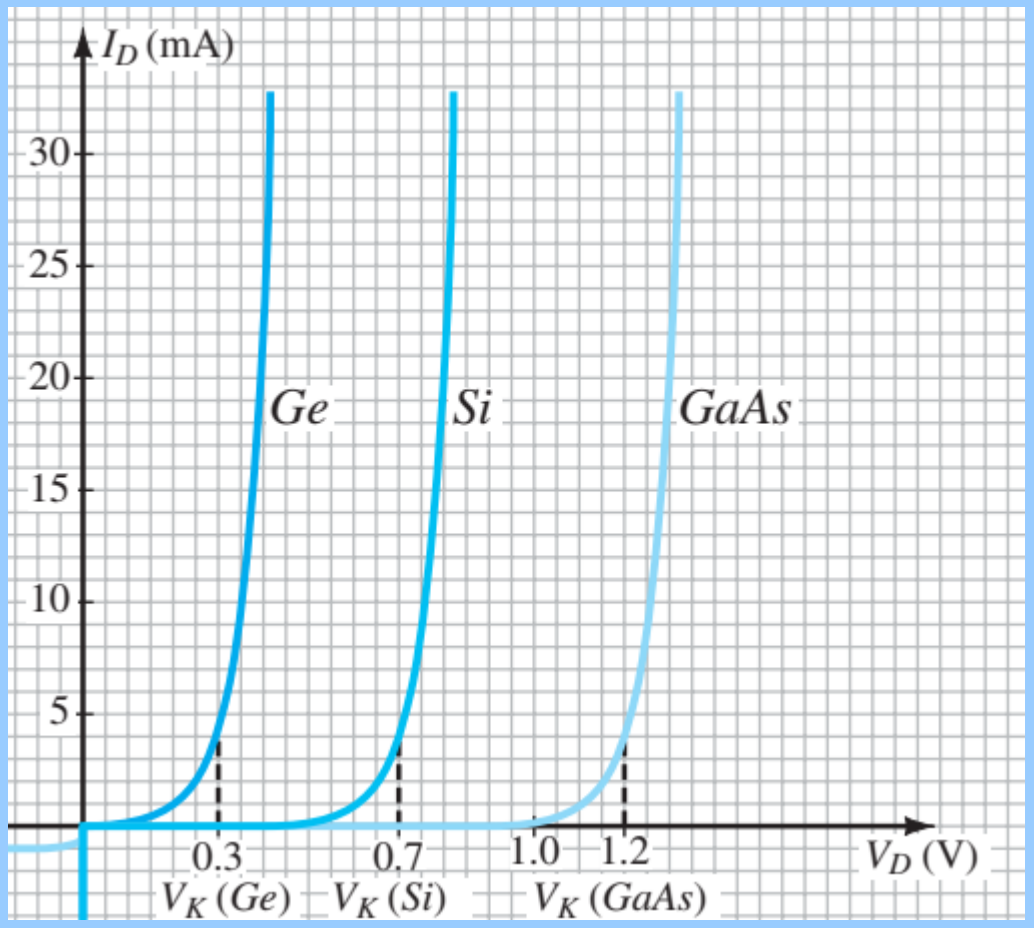
where I_s is the reverse saturation current
V_D is the applied forward-bias voltage across the diode
n is an ideality factor, which is a function of the operating conditions and physical construction; it has a range between 1 and 2 depending on a wide variety of factors ($n = 1$ will be assumed throughout this text unless otherwise noted).

The voltage V_T in Eq. (1.1) is called the *thermal voltage* and is determined by

$$V_T = \frac{kT_K}{q} \quad \text{(V)} \tag{1.3}$$

where k is Boltzmann's constant $= 1.38 \times 10^{-23}$ J/K
T_K is the absolute temperature in kelvins = 273 + the temperature in °C
q is the magnitude of electronic charge $= 1.6 \times 10^{-19}$ C

Actual Diode Characteristics



Load Line Analysis

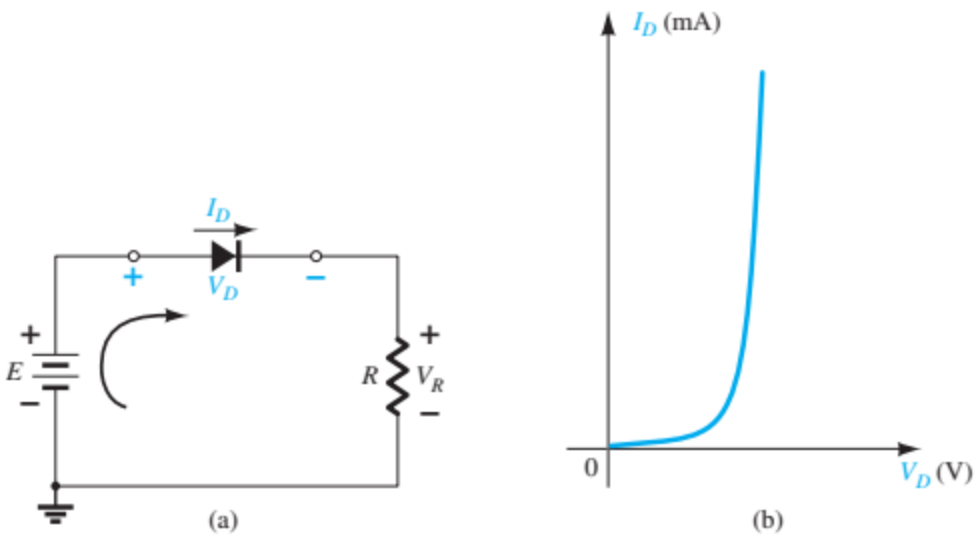


FIG. 2.1

Series diode configuration: (a) circuit; (b) characteristics.

Load Line Analysis

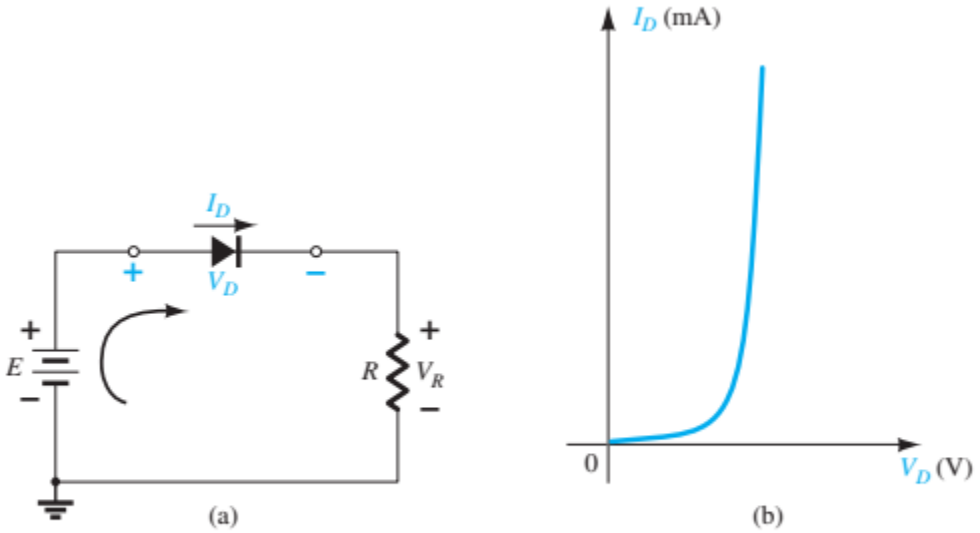


FIG. 2.1

Series diode configuration: (a) circuit; (b) characteristics.

$$I_D = I_s(e^{V_D/nV_T} - 1)$$

$$+E - V_D - V_R = 0$$

$$E = V_D + I_D R$$

Load Line Analysis

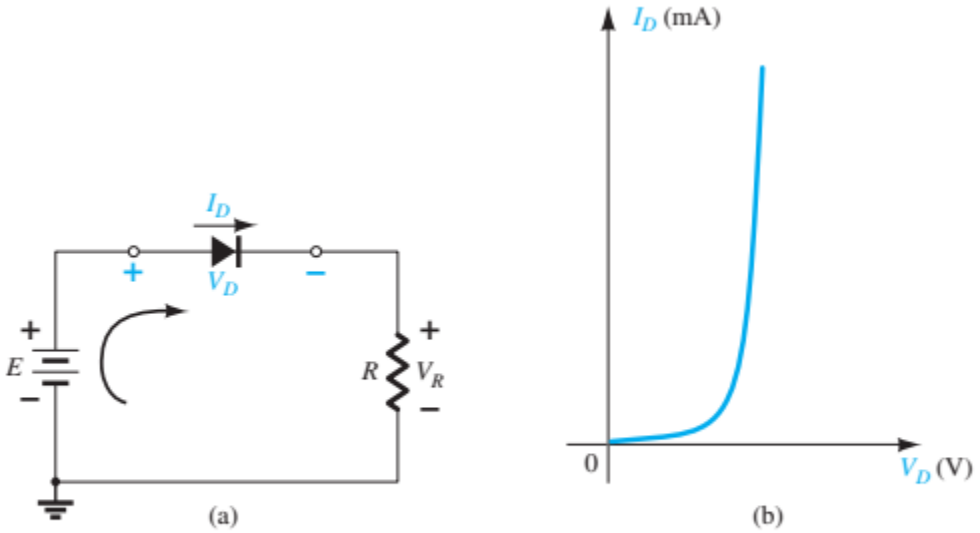


FIG. 2.1

Series diode configuration: (a) circuit; (b) characteristics.

$$I_D = I_s(e^{V_D/nV_T} - 1)$$

$$+E - V_D - V_R = 0$$

$$E = V_D + I_D R$$

$$I_D = \left.\frac{E}{R}\right|_{V_D=0\,\text{V}}$$

Load Line Analysis

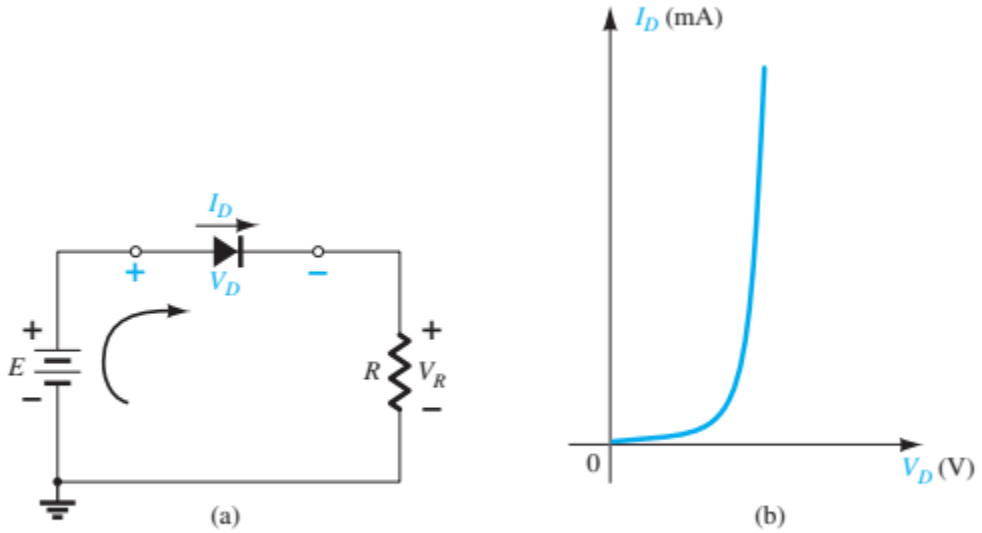


FIG. 2.1

Series diode configuration: (a) circuit; (b) characteristics.

$$I_D = I_s(e^{V_D/nV_T} - 1)$$

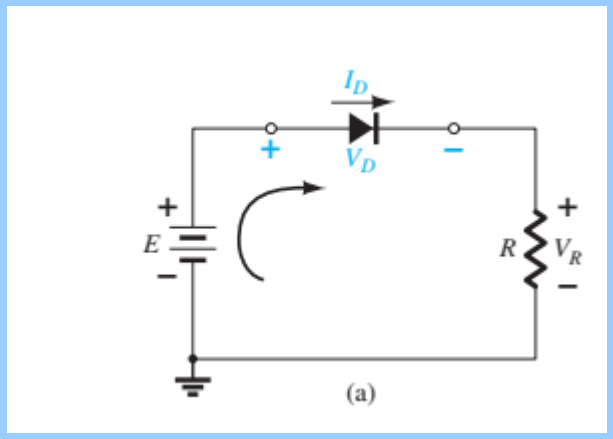
$$+E - V_D - V_R = 0$$

$$E = V_D + I_D R$$

$$I_D = \left.\frac{E}{R}\right|_{V_D=0\,\text{V}}$$

$$V_D = E\big|_{I_D=0\,\text{A}}$$

Load Line Analysis



$$I_D = I_s(e^{V_D/nV_T} - 1)$$

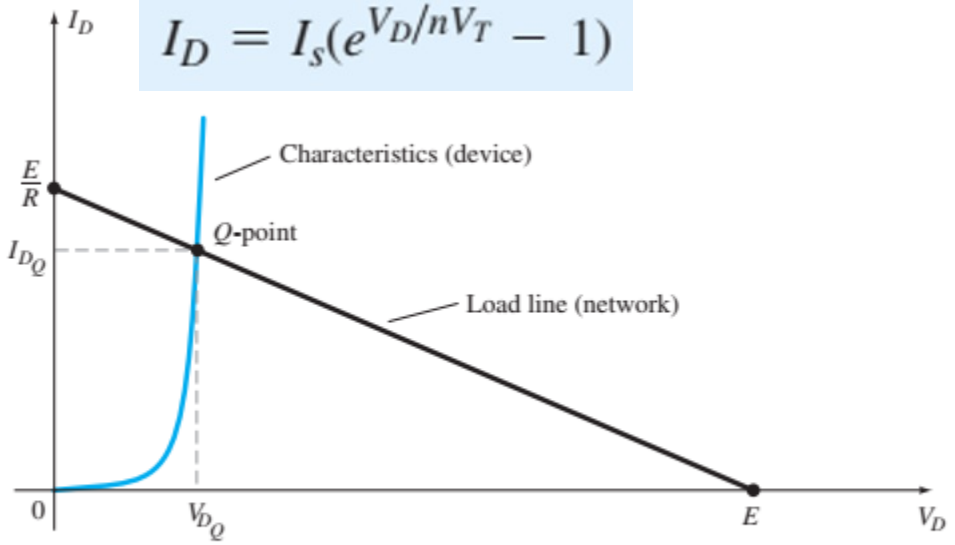


FIG. 2.2

Drawing the load line and finding the point of operation.

$$+E - V_D - V_R = 0$$

$$E = V_D + I_D R$$

$$I_D = \frac{E}{R}\bigg|_{V_D = 0\text{ V}}$$

$$V_D = E\big|_{I_D = 0\text{ A}}$$

Load Line Analysis

EXAMPLE 2.1 For the series diode configuration of Fig. 2.3a, employing the diode characteristics of Fig. 2.3b, determine:

a. V_{D_Q} and I_{D_Q}.
b. V_R.

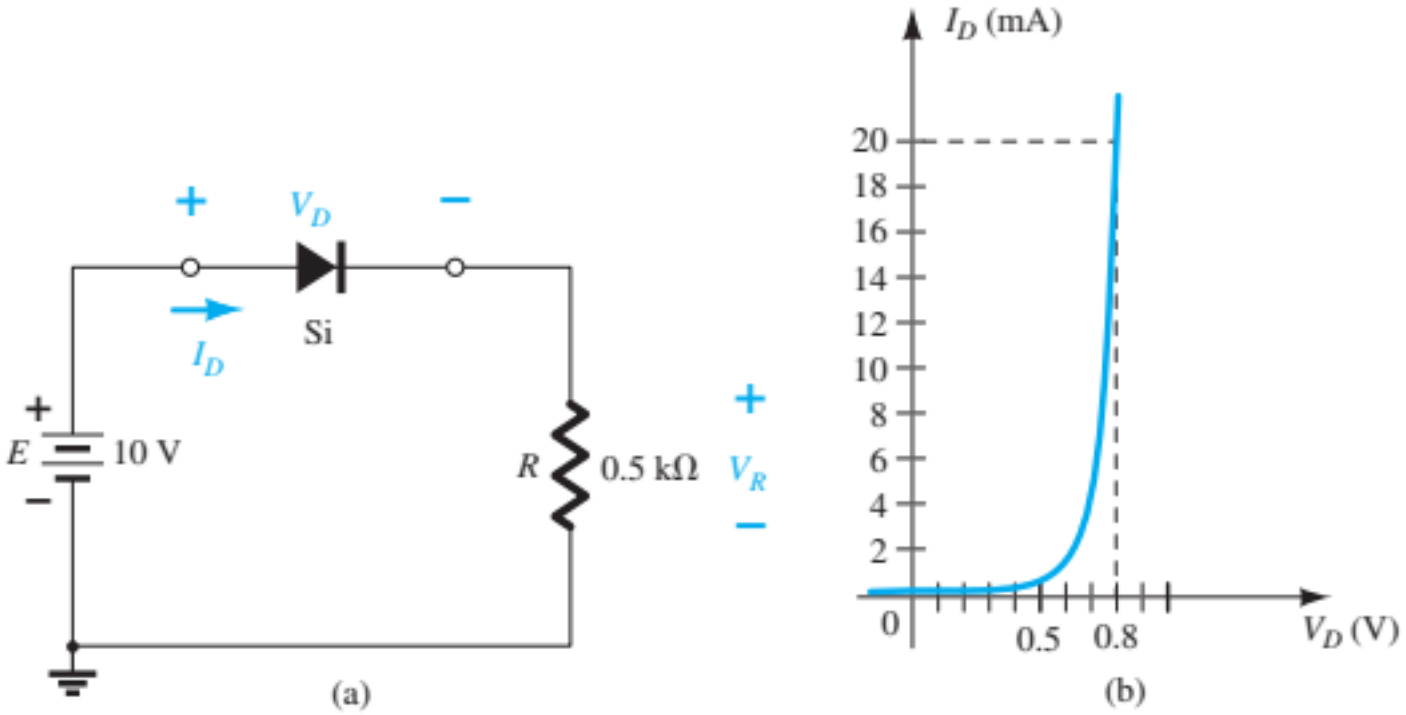


FIG. 2.3
(a) Circuit; (b) characteristics.

Load Line Analysis

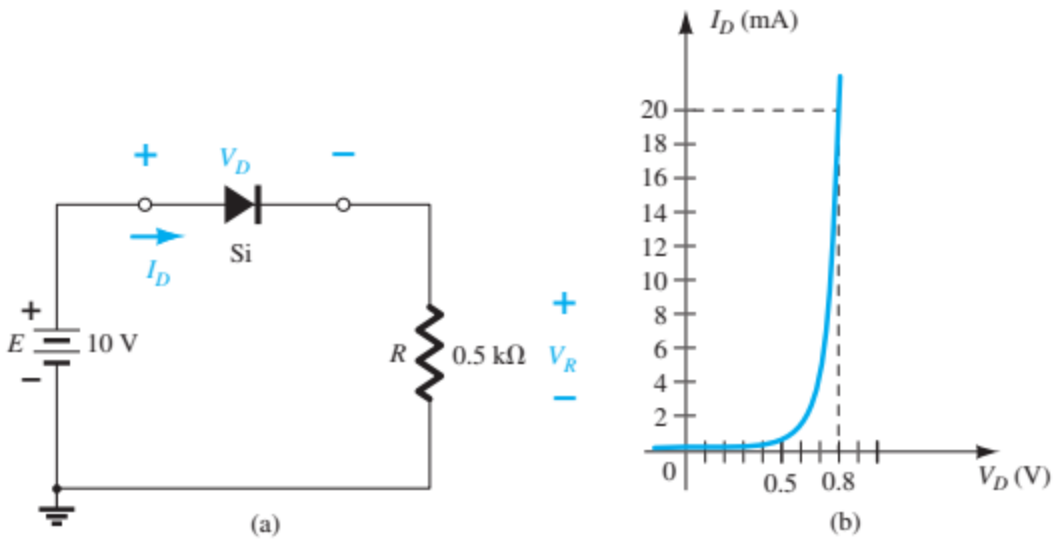


FIG. 2.3

(a) Circuit; (b) characteristics.

Solution:

a. Eq. (2.2): $$I_D = \left.\frac{E}{R}\right|_{V_D=0\text{ V}} = \frac{10\text{ V}}{0.5\text{ k}\Omega} = 20\text{ mA}$$

Eq. (2.3): $$V_D = E|_{I_D=0\text{ A}} = 10\text{ V}$$

Load Line Analysis

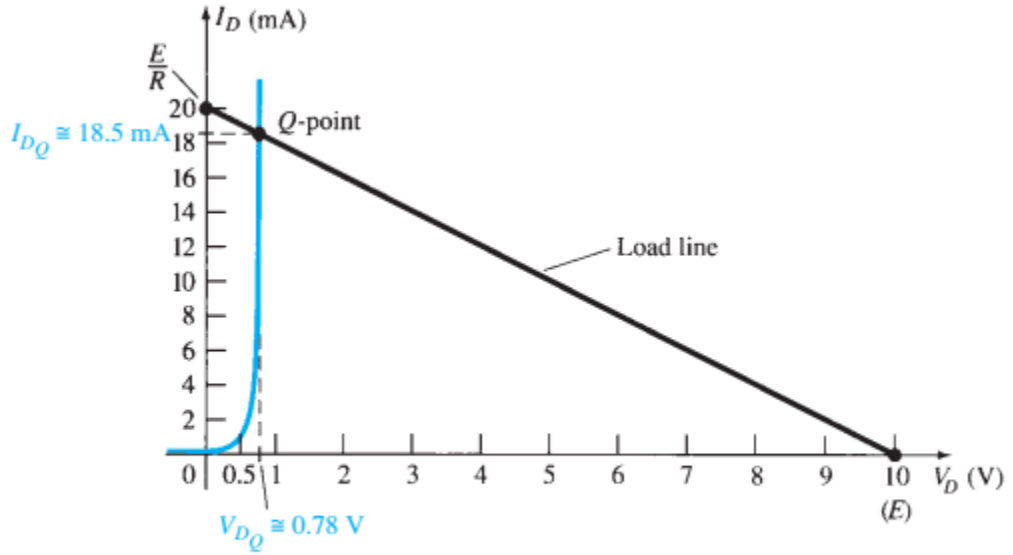
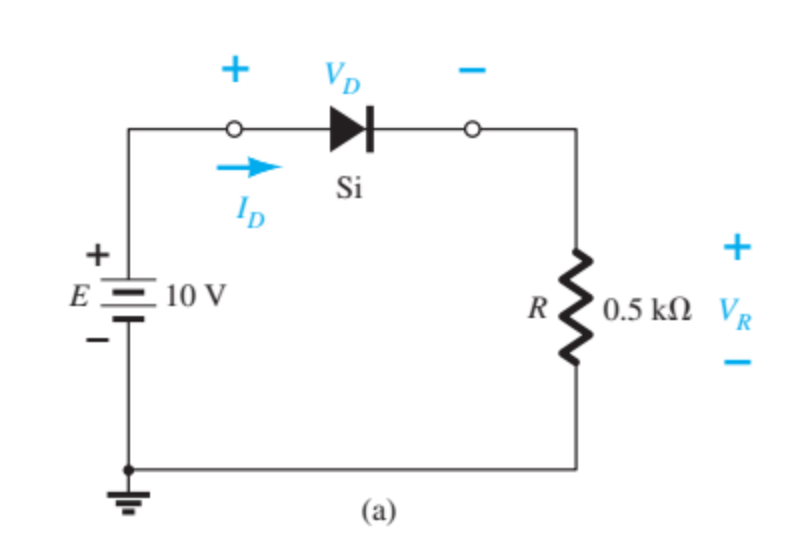


FIG. 2.4
Solution to Example 2.1.

$$I_D = \frac{E}{R}\bigg|_{V_D=0\text{ V}} = \frac{10\text{ V}}{0.5\text{ k}\Omega} = 20\text{ mA}$$
$$V_D = E|_{I_D=0\text{ A}} = 10\text{ V}$$

$$V_{D_Q} \cong \mathbf{0.78\text{ V}}$$
$$I_{D_Q} \cong \mathbf{18.5\text{ mA}}$$

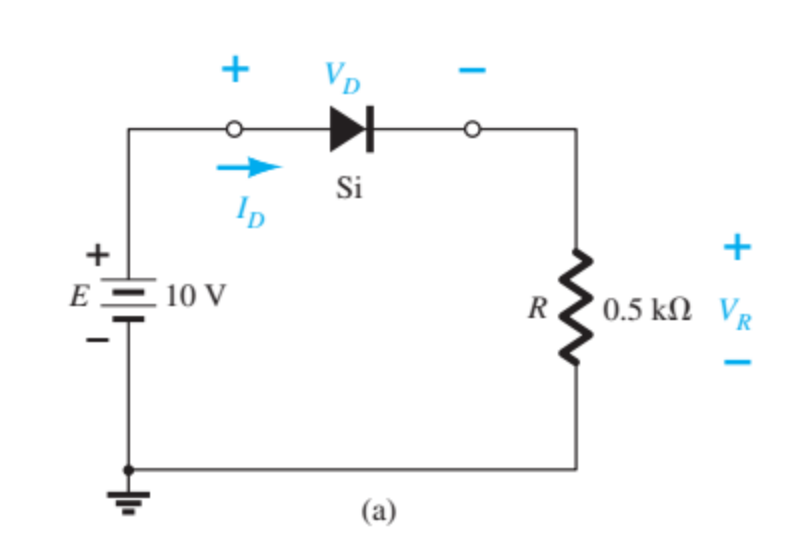
Load Line Analysis



$V_{D_Q} \cong$ **0.78 V**
$I_{D_Q} \cong$ **18.5 mA**

V_R ?

Load Line Analysis



$V_{D_Q} \cong$ **0.78 V**
$I_{D_Q} \cong$ **18.5 mA**

V_R ?

$$V_R = E - V_D = 10\,\text{V} - 0.78\,\text{V} = \mathbf{9.22\,V}$$

Actual Diode Characteristics

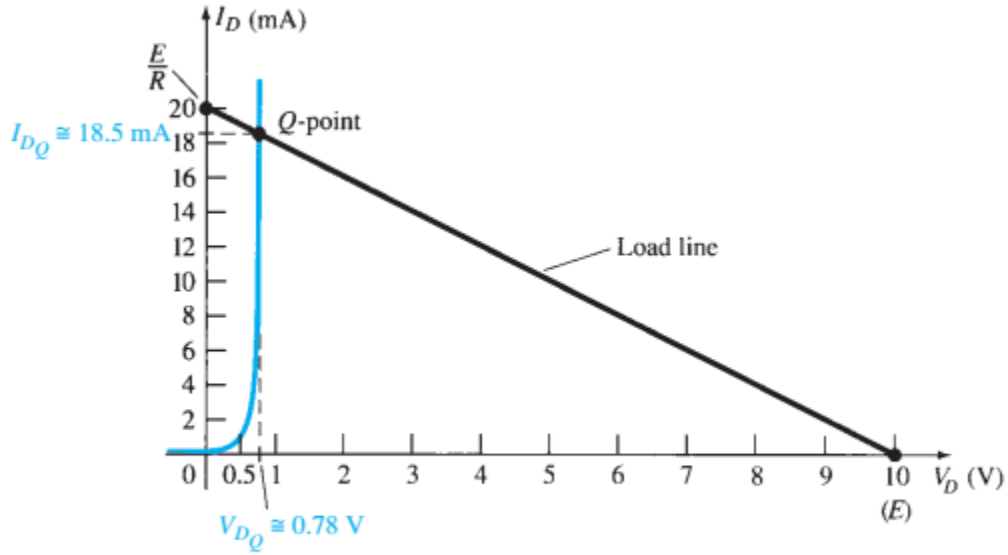
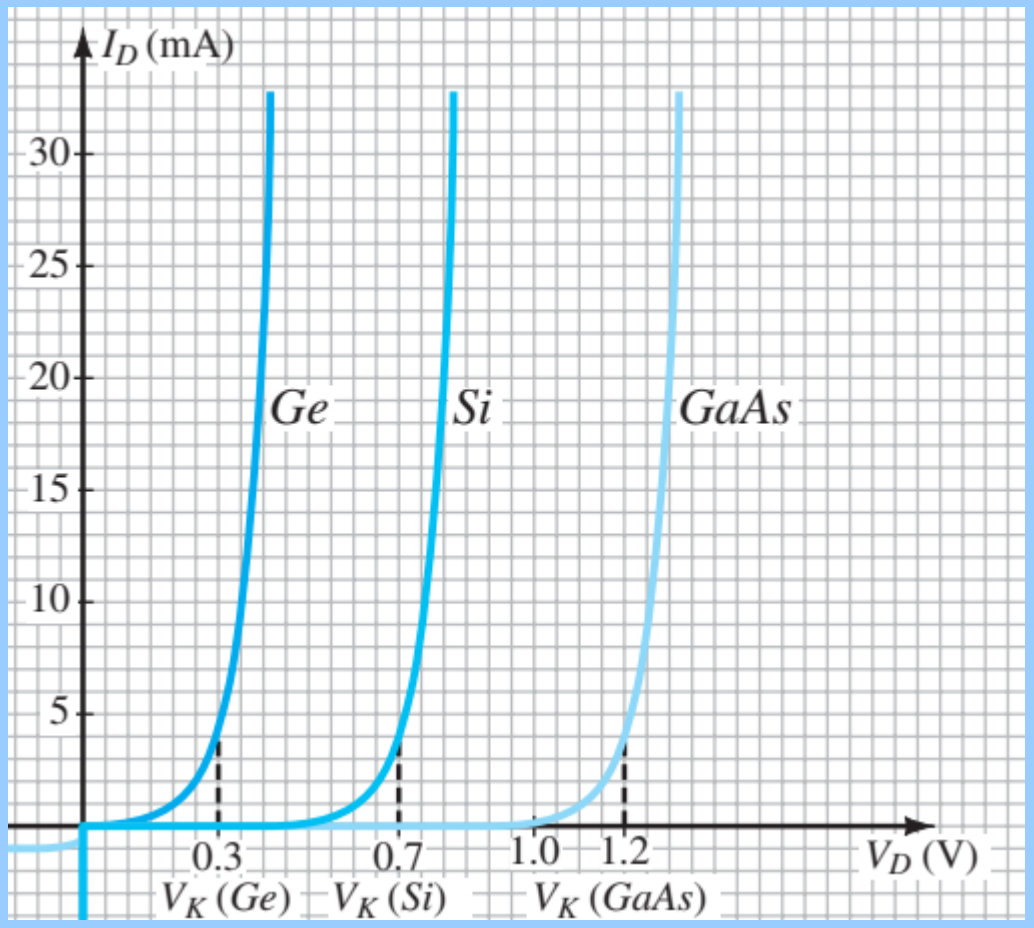


FIG. 2.4

Solution to Example 2.1.

Actual Diode Characteristics



Forward Bias Voltage

The point at which the diode changes from no-bias condition to forward-bias condition occurs when the electrons and holes are given sufficient energy to cross the *p-n* junction. This energy comes from the external voltage applied across the diode.

The forward bias voltage required for a:

- **gallium arsenide diode ≅ 1.2 V**
- **silicon diode ≅ 0.7 V**
- **germanium diode ≅ 0.3 V**

Diode – Ideal vs. Actual

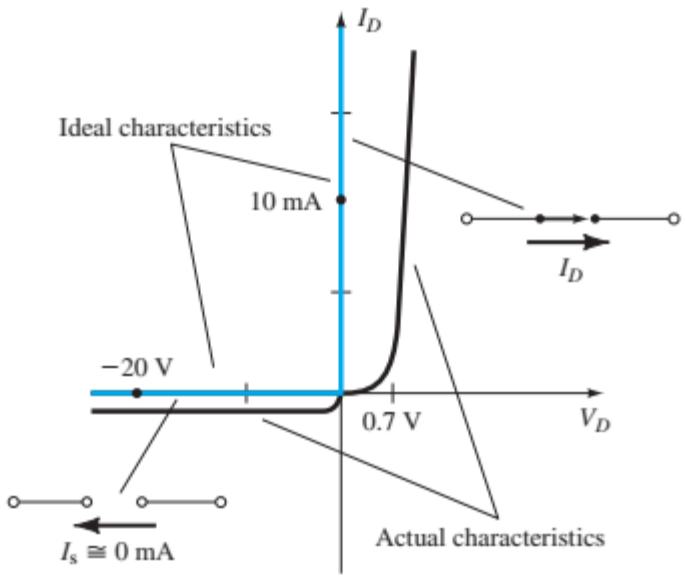


FIG. 1.22
Ideal versus actual semiconductor characteristics.

Ideal Diode

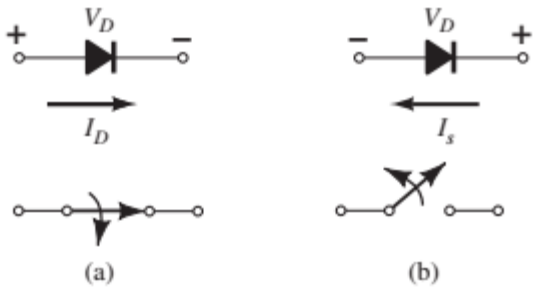


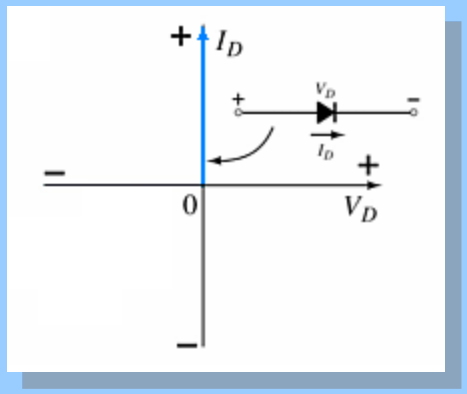
FIG. 1.21
Ideal semiconductor diode: (a) forward-biased; (b) reverse-biased.

The diode is a 2-terminal device.

A diode ideally conducts in only one direction.

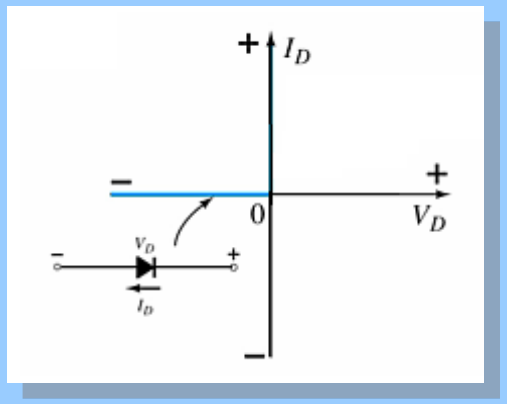
Ideal Diode

Conduction Region



- The voltage across the diode is 0 V
- The current is infinite
- The forward resistance is defined as $R_F = V_F / I_F$
- The diode acts like a short

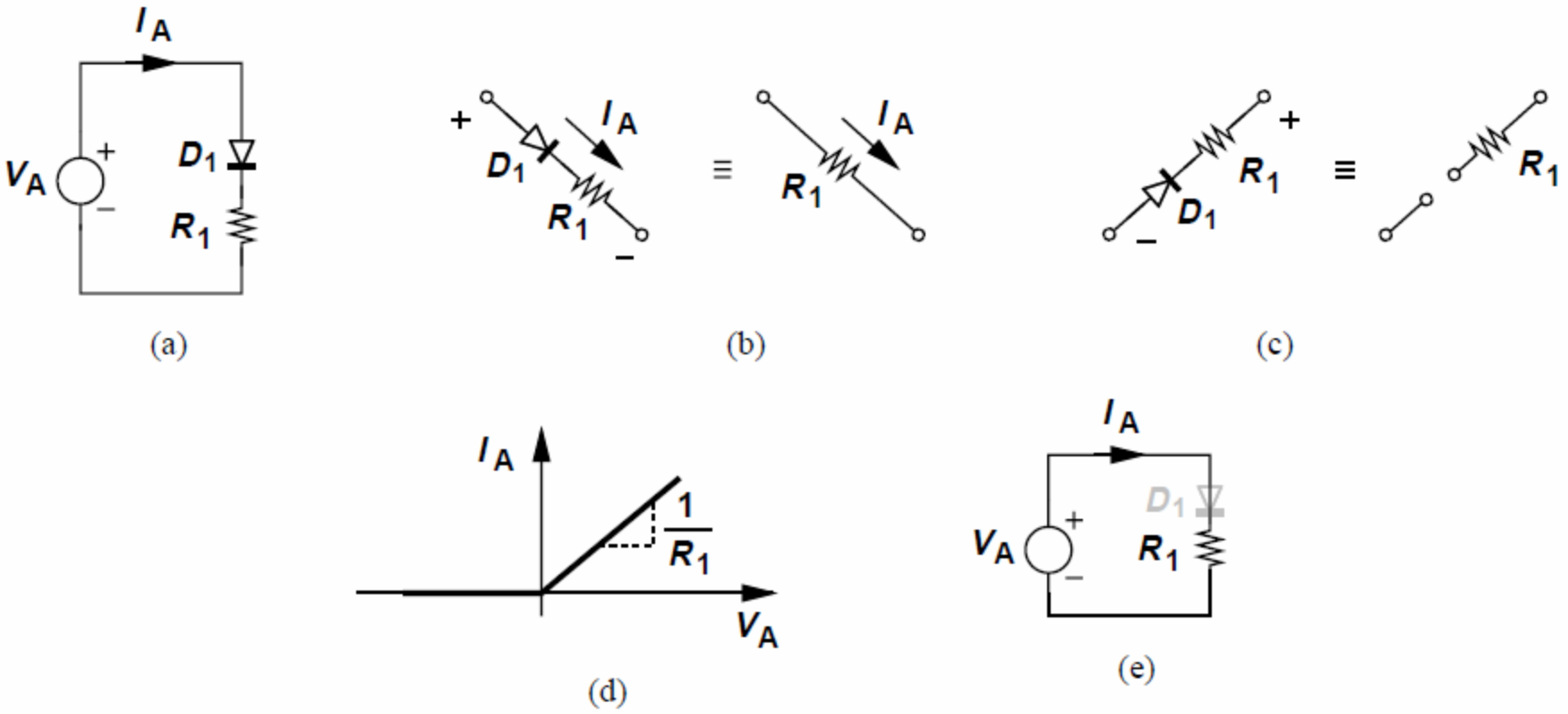
Non-Conduction Region



- All of the voltage is across the diode
- The current is 0 A
- The reverse resistance is defined as $R_R = V_R / I_R$
- The diode acts like open

EXAMPLE

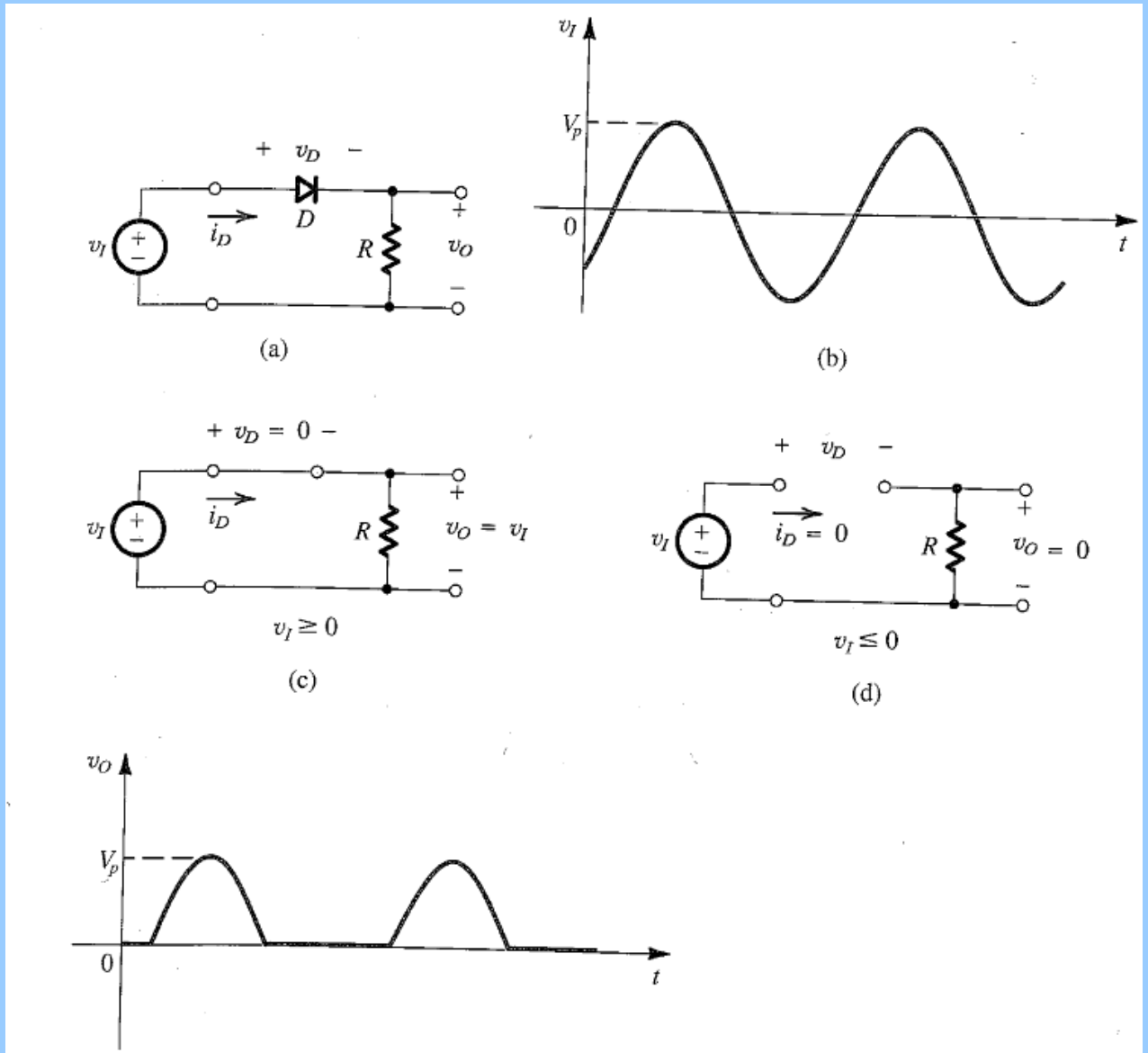
- Plot the I/V characteristic for the diode-resistor combination.



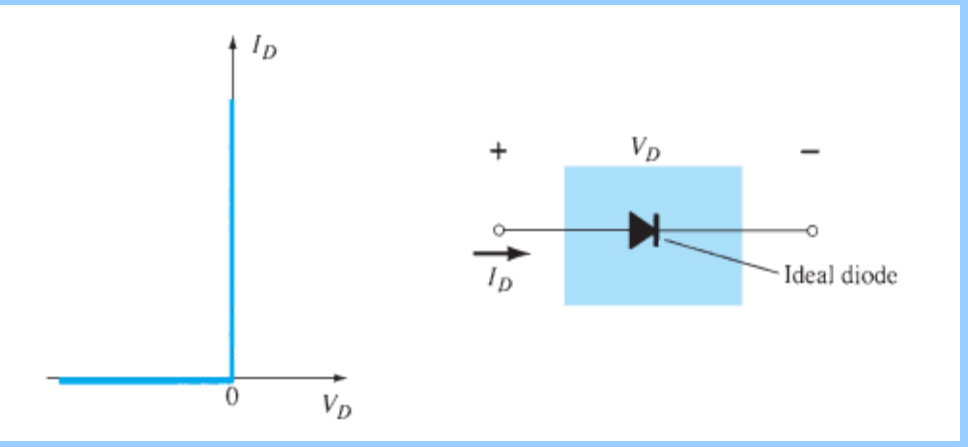
(a) Diode-resistor series combination, (b) equivalent circuit under forward bias, (c) equivalent circuit under reverse bias, (d) I/V characteristic, (e) equivalent circuit if D_1 is on.

EXAMPLE

- Plot the Vo for the diode-resistor combination.



Diode



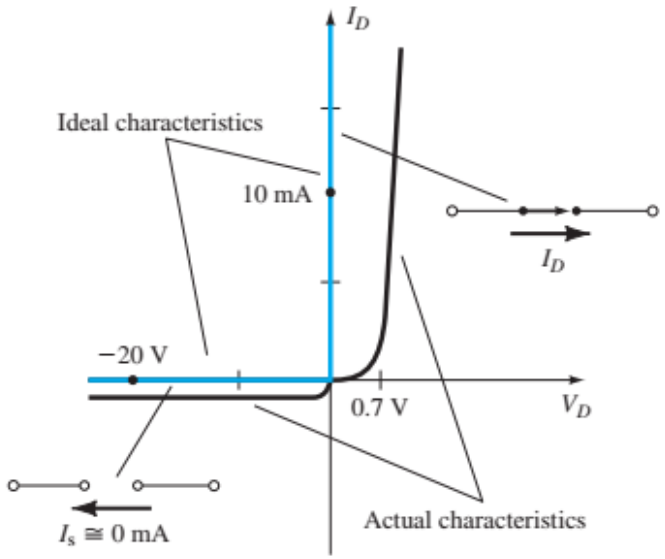
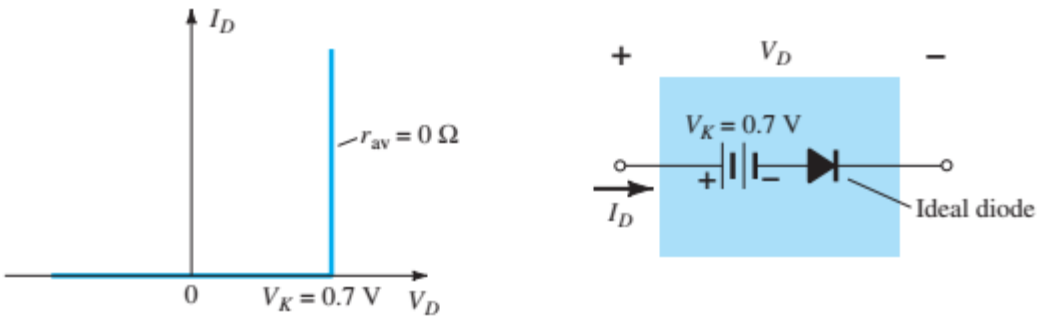


FIG. 1.22

Ideal versus actual semiconductor characteristics.

Diode



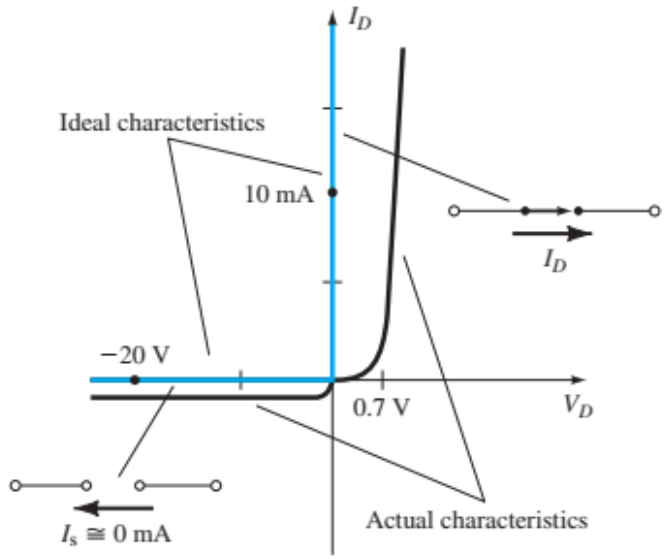


FIG. 1.22

Ideal versus actual semiconductor characteristics.

Diode

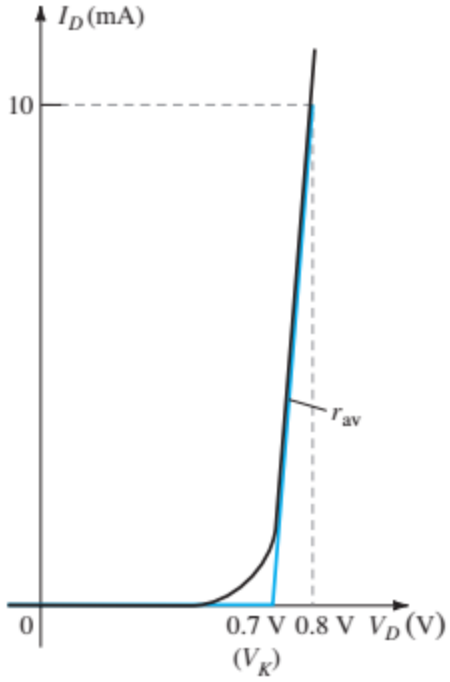
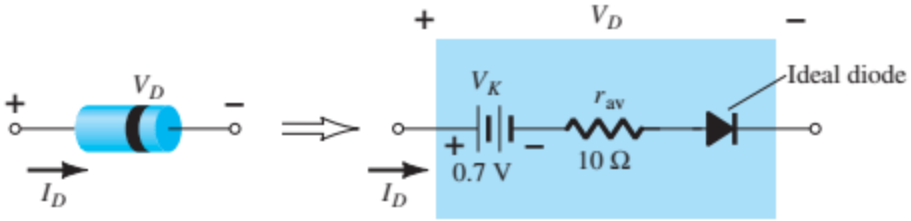


FIG. 1.29

Defining the piecewise-linear equivalent circuit using straight-line segments to approximate the characteristic curve.



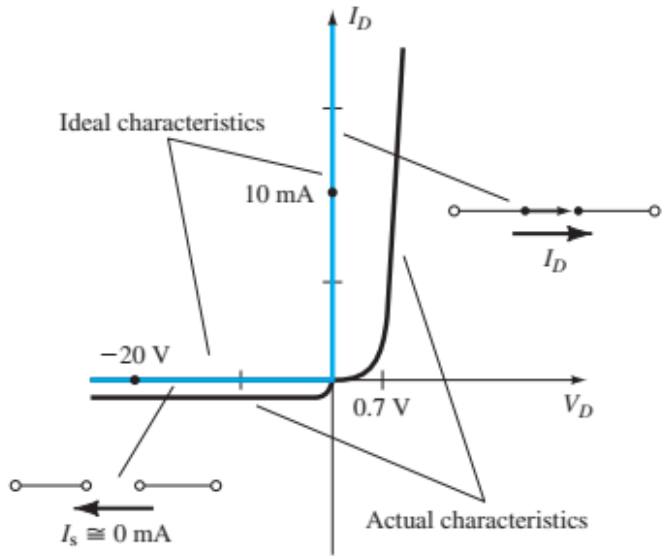


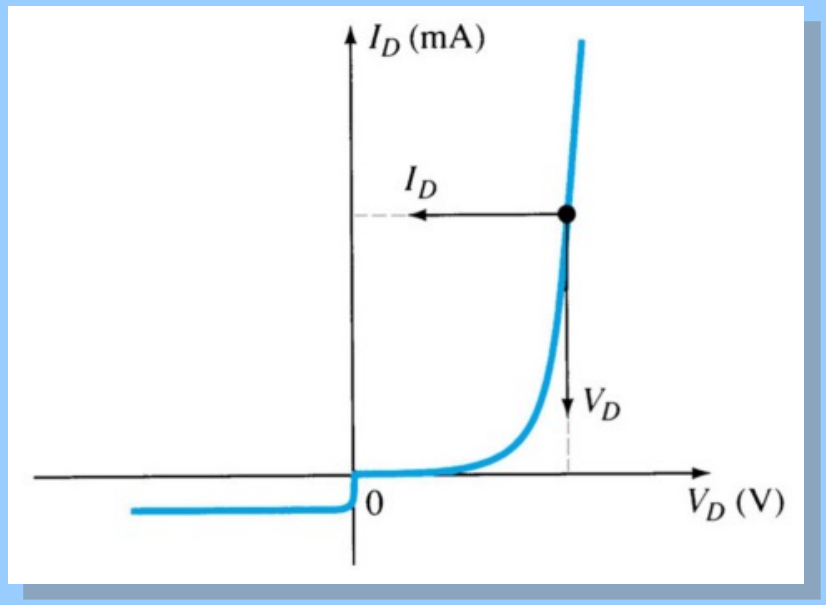
FIG. 1.22

Ideal versus actual semiconductor characteristics.

DC (Static) Resistance

For a specific applied DC voltage V_D, the diode has a specific current I_D, and a specific resistance R_D.

$$R_D = \frac{V_D}{I_D}$$



Diode – AC resistance

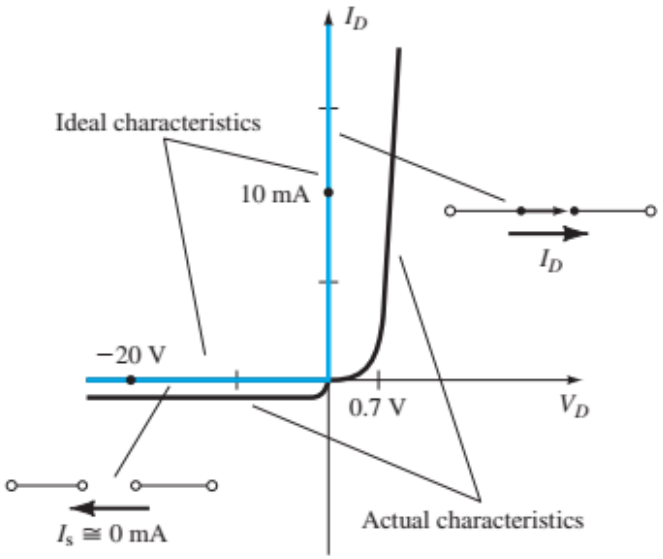
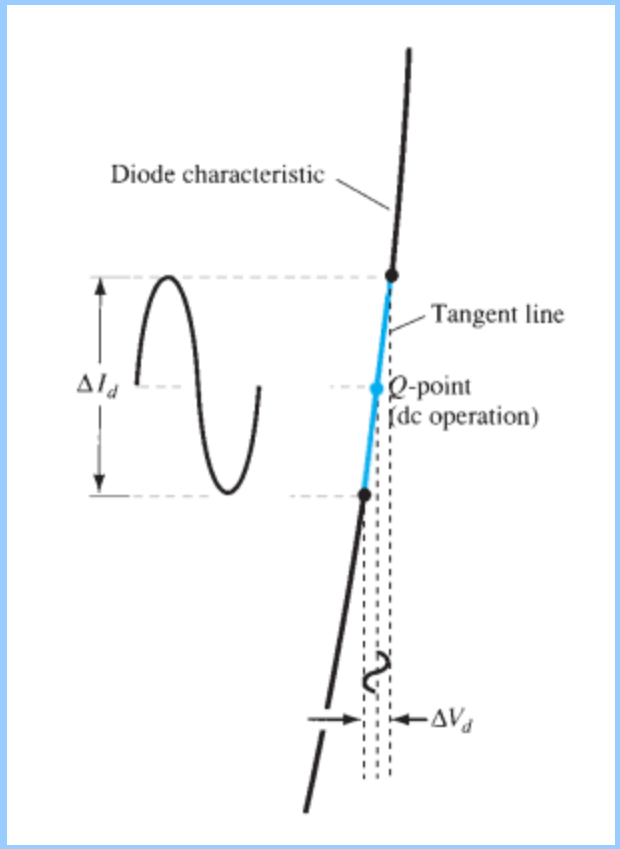
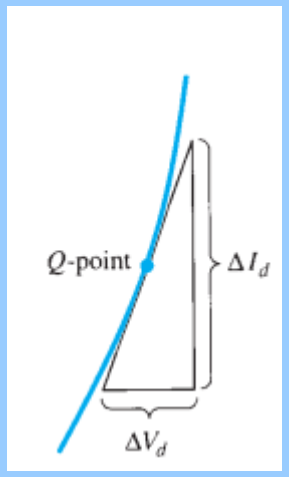


FIG. 1.22
Ideal versus actual semiconductor characteristics.



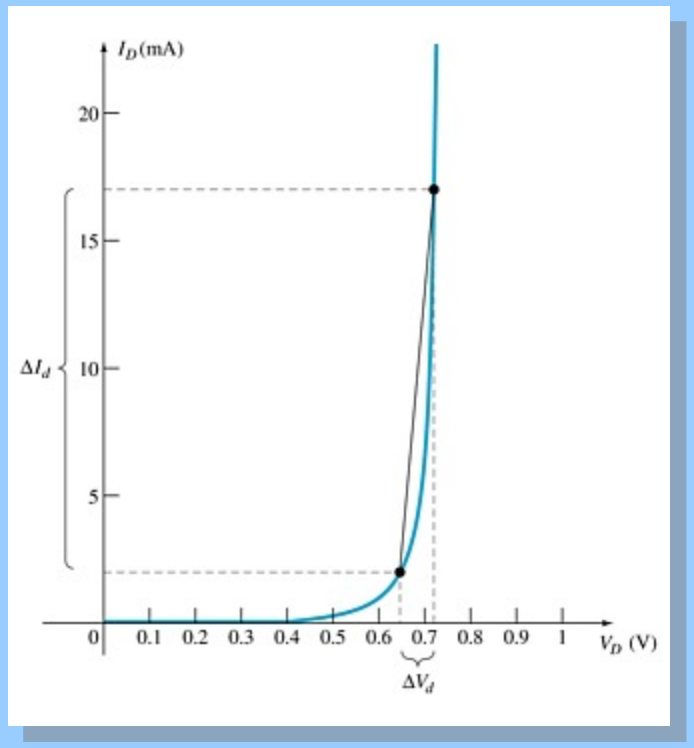
$$r_d = \frac{\Delta V_d}{\Delta I_d}$$



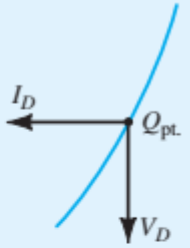
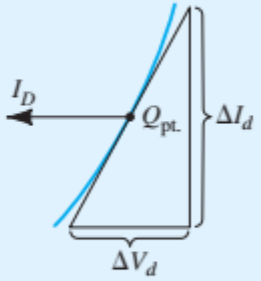
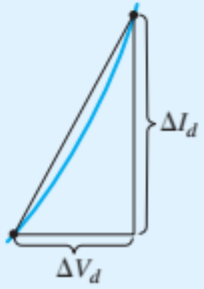
Average AC Resistance

$$r_{av} = \frac{\Delta V_d}{\Delta I_d} \Big| \text{ pt. to pt.}$$

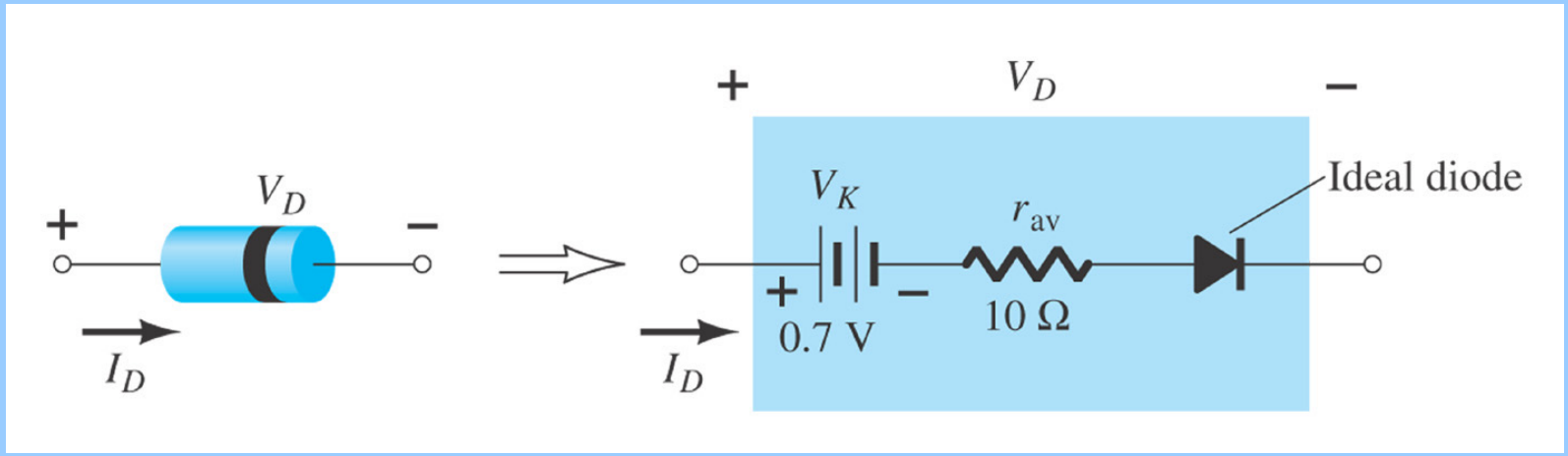
AC resistance can be calculated using the current and voltage values for two points on the diode characteristic curve.

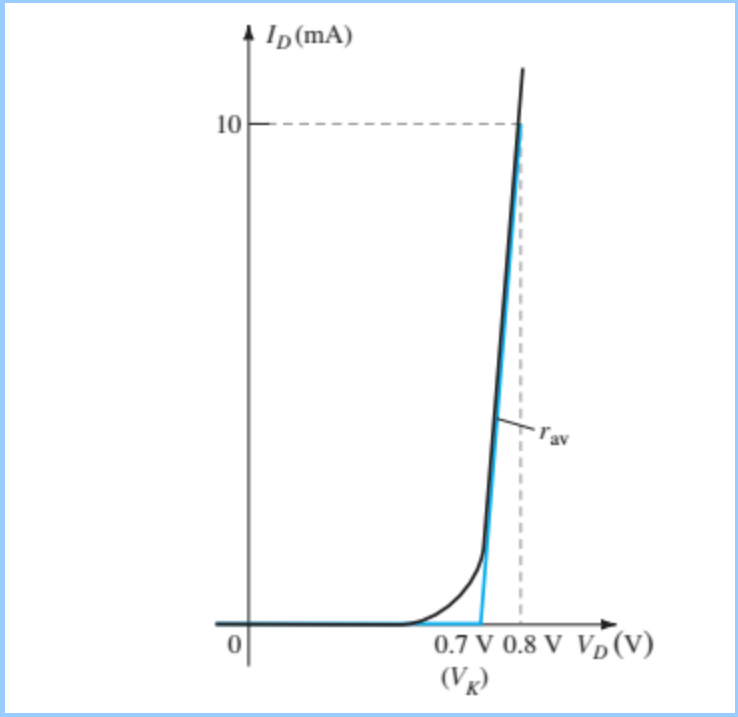


Diode – Resistance Levels

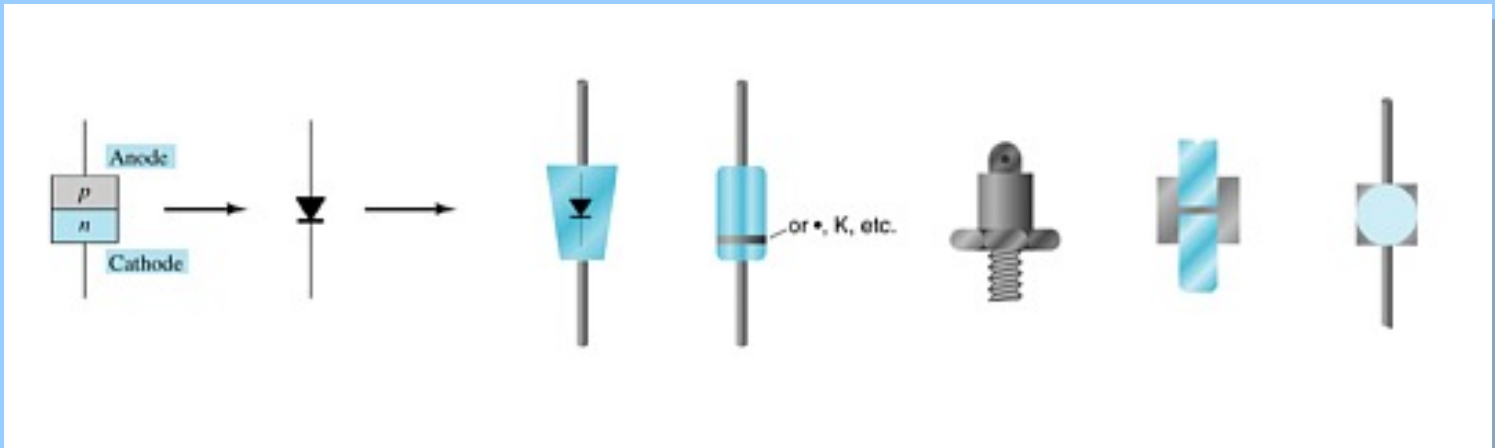
Type	Equation	Special Characteristics	Graphical Determination
DC or static	$R_D = \frac{V_D}{I_D}$	Defined as a point on the characteristics	
AC or dynamic	$r_d = \frac{\Delta V_d}{\Delta I_d} = \frac{26\text{ mV}}{I_D}$	Defined by a tangent line at the Q-point	
Average ac	$r_{av} = \left.\frac{\Delta V_d}{\Delta I_d}\right\|_{\text{pt. to pt.}}$	Defined by a straight line between limits of operation	

Diode Equivalent Circuit





Diode Symbol and Packaging



The anode is abbreviated A
The cathode is abbreviated K

Diode Checker

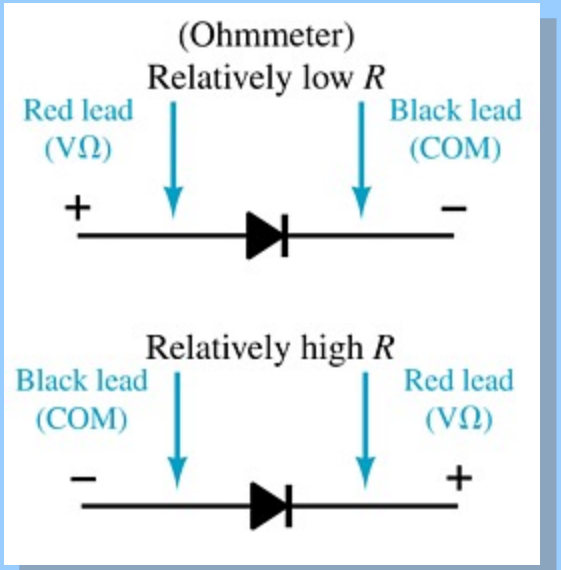
Many digital multimeters have a diode checking function.
The diode should be tested out of circuit.

A normal diode exhibits its forward voltage:

- **Gallium arsenide ≅ 1.2 V**
- **Silicon diode ≅ 0.7 V**
- **Germanium diode ≅ 0.3 V**

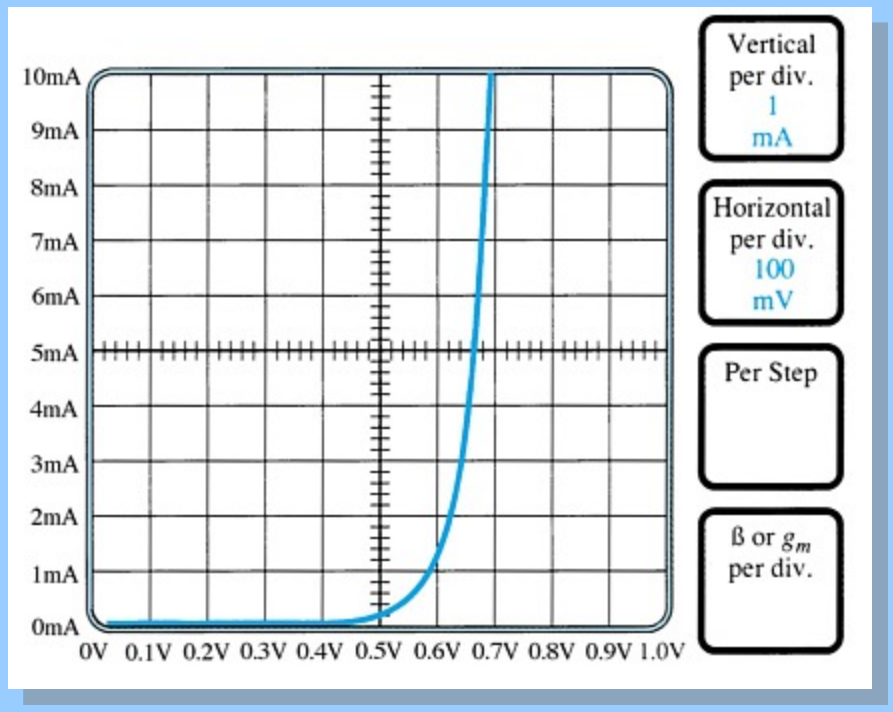
Ohmmeter

An ohmmeter set on a low Ohms scale can be used to test a diode. The diode should be tested out of circuit.



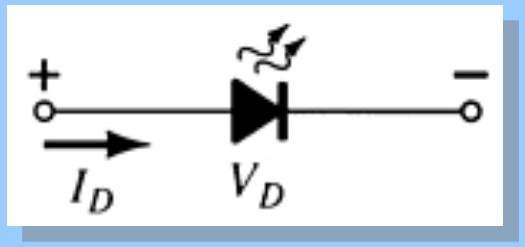
Curve Tracer

A curve tracer displays the characteristic curve of a diode in the test circuit. This curve can be compared to the specifications of the diode from a data sheet.



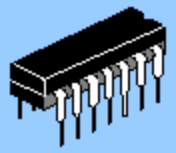
Light-Emitting Diode (LED)

An LED emits photons when it is forward biased.
These can be in the infrared or visible spectrum.
The forward bias voltage is usually in the range of 2 V to 3 V.



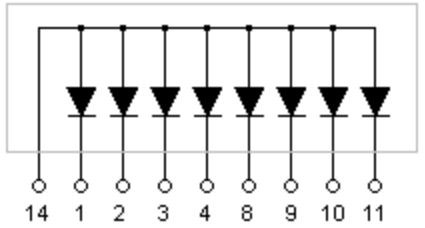
Diode Arrays

Multiple diodes can be packaged together in an integrated circuit (IC).

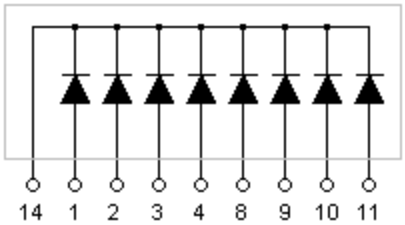


A variety of combinations exist.

Common Anode



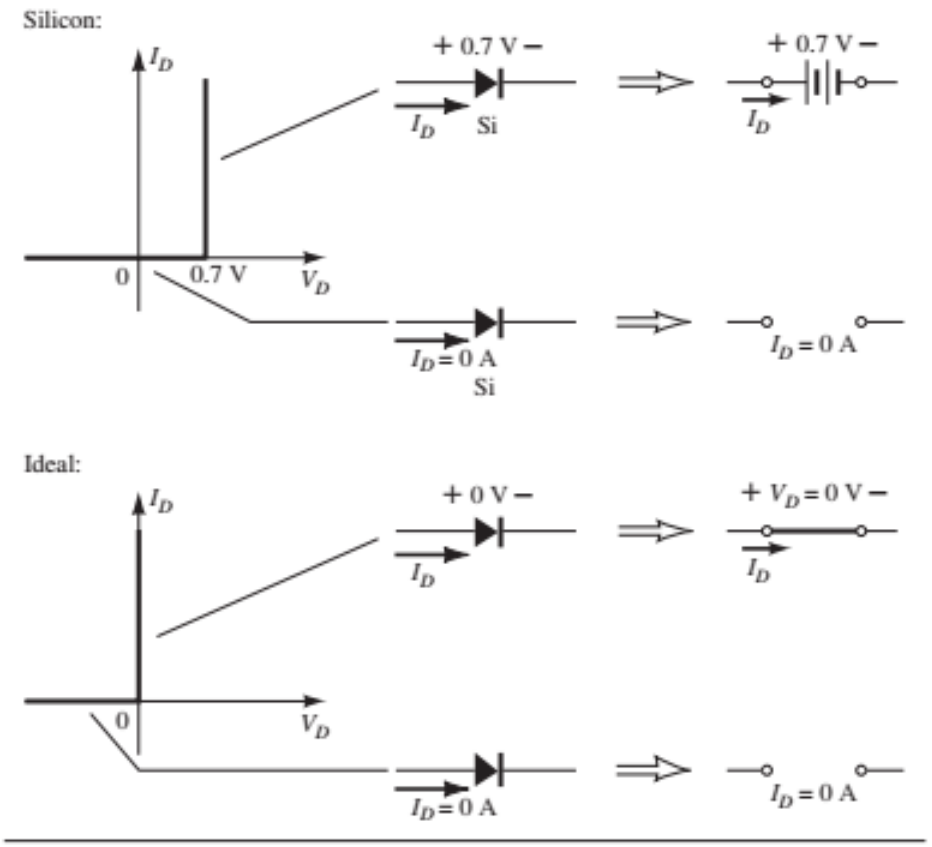
Common Cathode



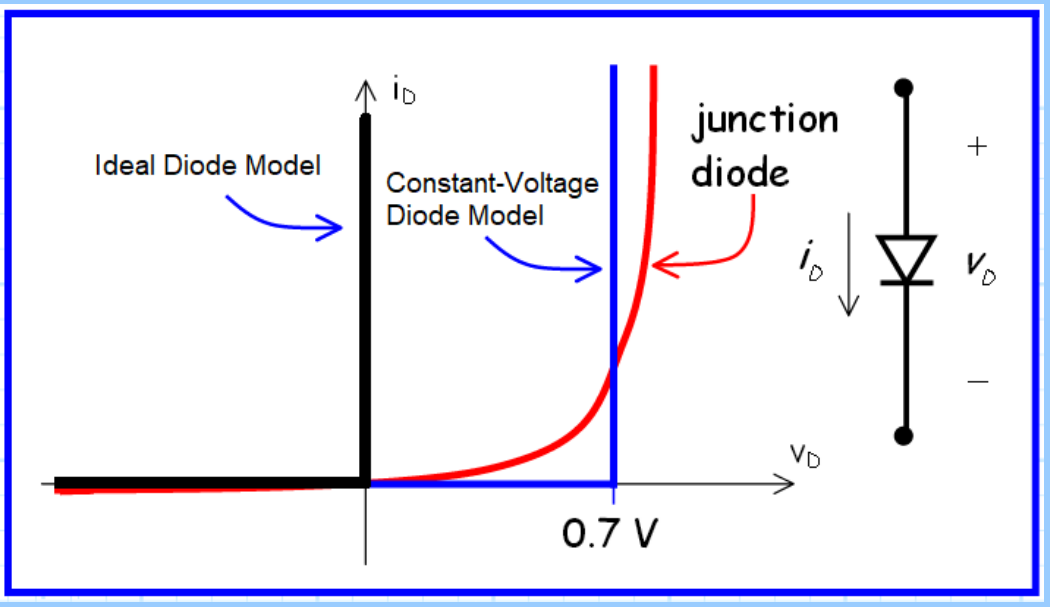
Constant Si Diode Model

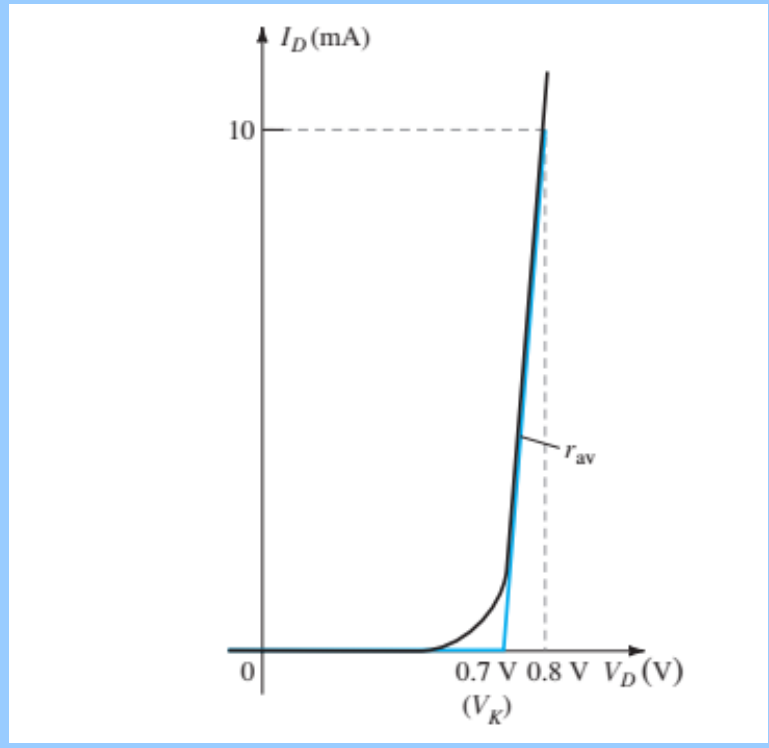
TABLE 2.1

Approximate and Ideal Semiconductor Diode Models.



Diode Models





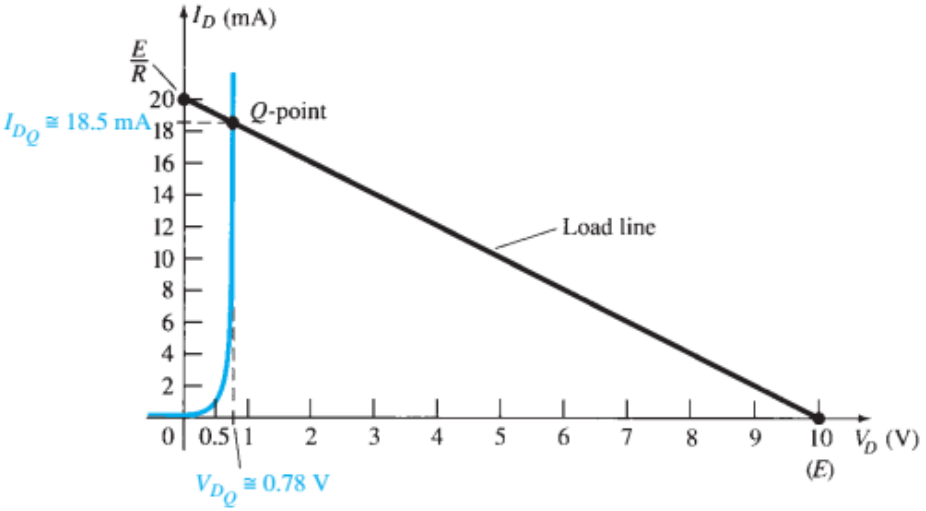


FIG. 2.4

Solution to Example 2.1.

Series Diode Configurations

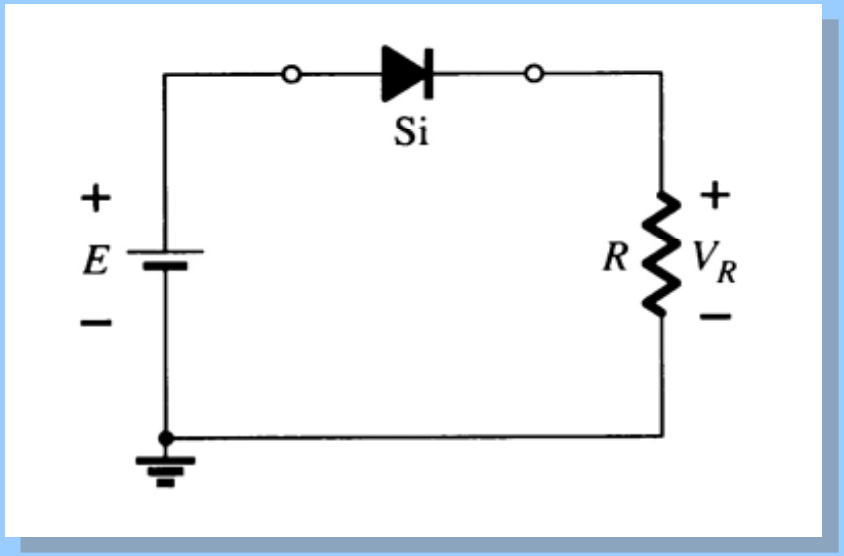
Forward Bias

Constants

- **Silicon Diode: $V_D = 0.7$ V**
- **Germanium Diode: $V_D = 0.3$ V**

Analysis (for silicon)

- $V_D = 0.7$ V (or $V_D = E$ if $E < 0.7$ V)
- $V_R = E - V_D$
- $I_D = I_R = I_T = V_R / R$



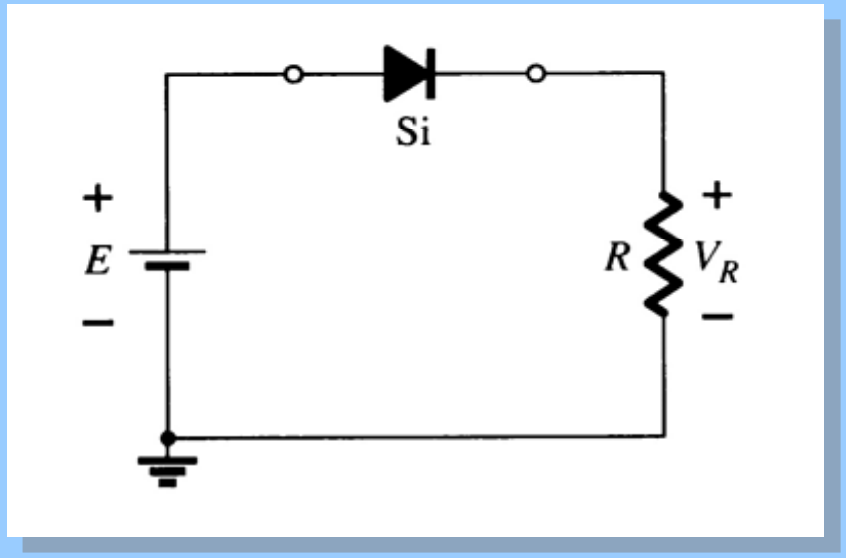
Series Diode Configurations

Reverse Bias

Diodes ideally behave as open circuits

Analysis

- $V_D = E$
- $V_R = 0\ \text{V}$
- $I_D = 0\ \text{A}$



Series Diode Configurations

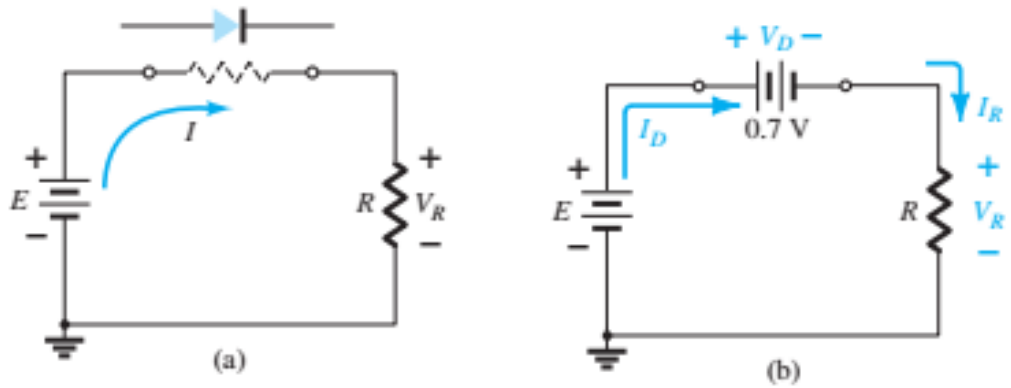


FIG. 2.9

(a) Determining the state of the diode of Fig. 2.8; (b) substituting the equivalent model for the "on" diode of Fig. 2.9a.

Series Diode Configurations

EXAMPLE 2.4 For the series diode configuration of Fig. 2.13, determine V_D, V_R, and I_D.

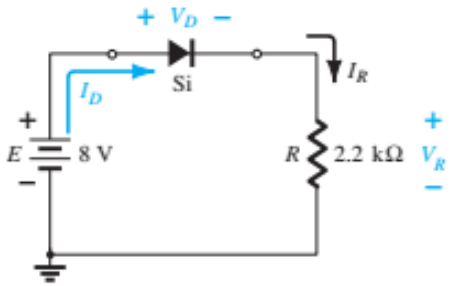


FIG. 2.13
Circuit for Example 2.4.

Series Diode Configurations

EXAMPLE 2.4 For the series diode configuration of Fig. 2.13, determine V_D, V_R, and I_D.

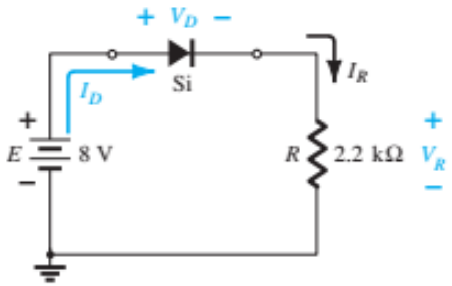


FIG. 2.13
Circuit for Example 2.4.

Solution: Since the applied voltage establishes a current in the clockwise direction to match the arrow of the symbol and the diode is in the "on" state,

$$V_D = \mathbf{0.7\ V}$$

$$V_R = E - V_D = 8\ \text{V} - 0.7\ \text{V} = \mathbf{7.3\ V}$$

$$I_D = I_R = \frac{V_R}{R} = \frac{7.3\ \text{V}}{2.2\ \text{k}\Omega} \cong \mathbf{3.32\ mA}$$

Series Diode Configurations

EXAMPLE 2.5 Repeat Example 2.4 with the diode reversed.

Solution: Removing the diode, we find that the direction of I is opposite to the arrow in the diode symbol and the diode equivalent is the open circuit no matter which model is employed. The result is the network of Fig. 2.14, where $I_D = \mathbf{0\ A}$ due to the open circuit. Since $V_R = I_R R$, we have $V_R = (0)R = \mathbf{0\ V}$. Applying Kirchhoff's voltage law around the closed loop yields

$$E - V_D - V_R = 0$$

and

$$V_D = E - V_R = E - 0 = E = 8\ \text{V}$$

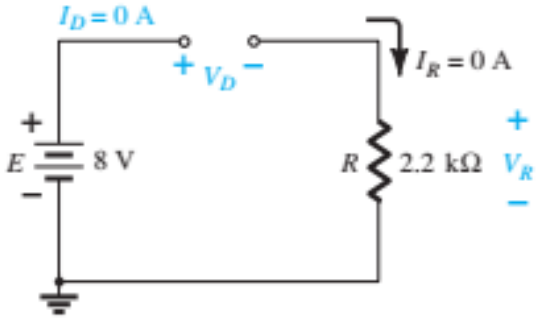


FIG. 2.14
Determining the unknown quantities for Example 2.5.

Series Diode Configurations

EXAMPLE 2.7 Determine V_o and I_D for the series circuit of Fig. 2.19.

Solution: An attack similar to that applied in Example 2.4 will reveal that the resulting current has the same direction as the arrowheads of the symbols of both diodes, and the network of Fig. 2.20 results because $E = 12\text{ V} > (0.7\text{ V} + 1.8\text{ V}\ [\text{Table 1.8}]) = 2.5\text{ V}$. Note the redrawn supply of 12 V and the polarity of V_o across the 680-Ω resistor. The resulting voltage is

$$V_o = E - V_{K_1} - V_{K_2} = 12\text{ V} - 2.5\text{ V} = \mathbf{9.5\text{ V}}$$

and

$$I_D = I_R = \frac{V_R}{R} = \frac{V_o}{R} = \frac{9.5\text{ V}}{680\ \Omega} = \mathbf{13.97\text{ mA}}$$

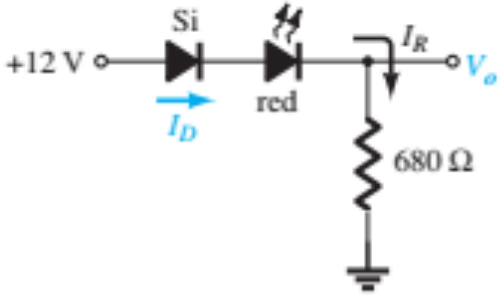


FIG. 2.19
Circuit for Example 2.7.

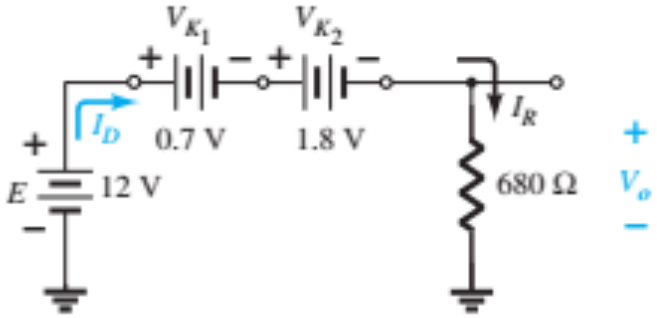


FIG. 2.20
Determining the unknown quantities for Example 2.7.

Series Diode Configurations

EXAMPLE 2.8 Determine I_D, V_{D_2}, and V_o for the circuit of Fig. 2.21.

Solution: Removing the diodes and determining the direction of the resulting current I result in the circuit of Fig. 2.22. There is a match in current direction for one silicon diode but not for the other silicon diode. The combination of a short circuit in series with an open circuit always results in an open circuit and $I_D = \mathbf{0\ A}$, as shown in Fig. 2.23.

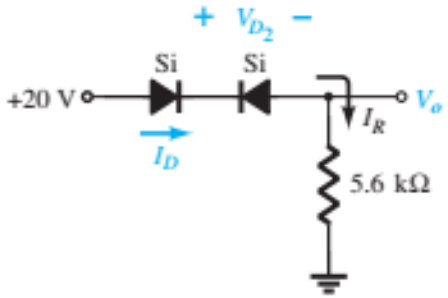


FIG. 2.21
Circuit for Example 2.8.

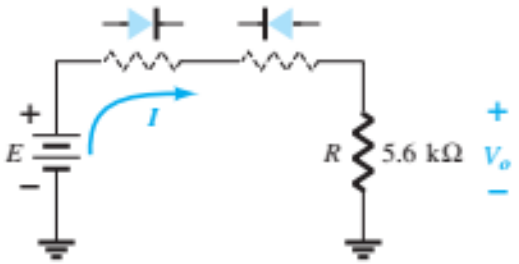


FIG. 2.22
Determining the state of the diodes of Fig. 2.21.

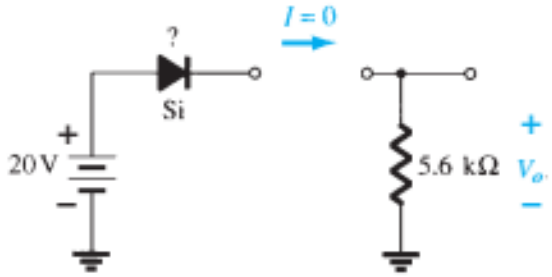


FIG. 2.23
Substituting the equivalent state for the open diode.

Series Diode Configurations

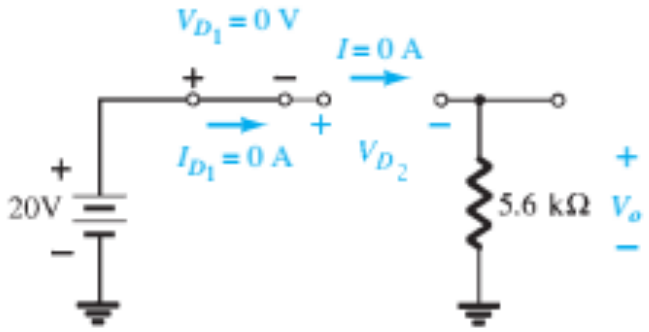


FIG. 2.24
Determining the unknown quantities for the circuit of Example 2.8.

The question remains as to what to substitute for the silicon diode. For the analysis to follow in this and succeeding chapters, simply recall for the actual practical diode that when $I_D = 0$ A, $V_D = 0$ V (and vice versa), as described for the no-bias situation in Chapter 1. The conditions described by $I_D = 0$ A and $V_{D_1} = 0$ V are indicated in Fig. 2.24. We have

$$V_o = I_R R = I_D R = (0\text{ A})R = \mathbf{0\text{ V}}$$

and

$$V_{D_2} = V_{\text{open circuit}} = E = \mathbf{20\text{ V}}$$

Applying Kirchhoff's voltage law in a clockwise direction gives

$$E - V_{D_1} - V_{D_2} - V_o = 0$$

and

$$V_{D_2} = E - V_{D_1} - V_o = 20\text{ V} - 0 - 0$$
$$= \mathbf{20\text{ V}}$$

with

$$V_o = \mathbf{0\text{ V}}$$

Series Diode Configurations

EXAMPLE 2.9 Determine I, V_1, V_2, and V_o for the series dc configuration of Fig. 2.25.

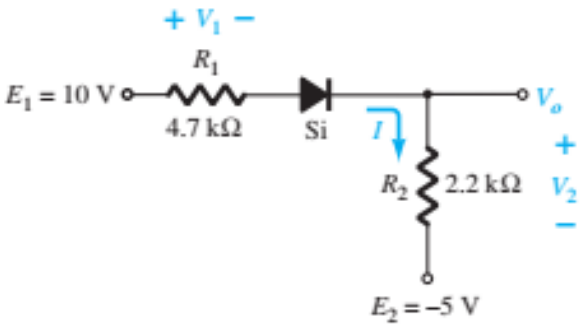


FIG. 2.25
Circuit for Example 2.9.

Series Diode Configurations

EXAMPLE 2.9 Determine I, V_1, V_2, and V_o for the series dc configuration of Fig. 2.25.

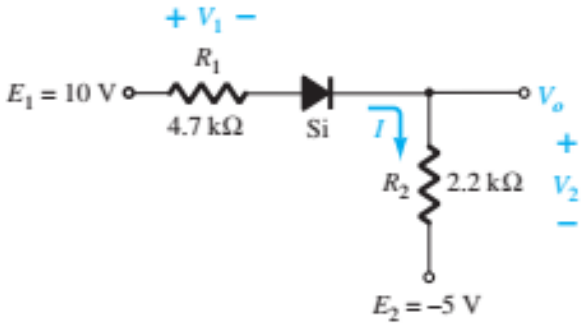
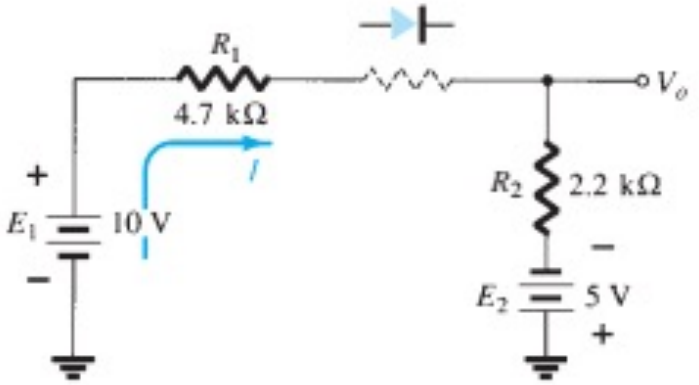
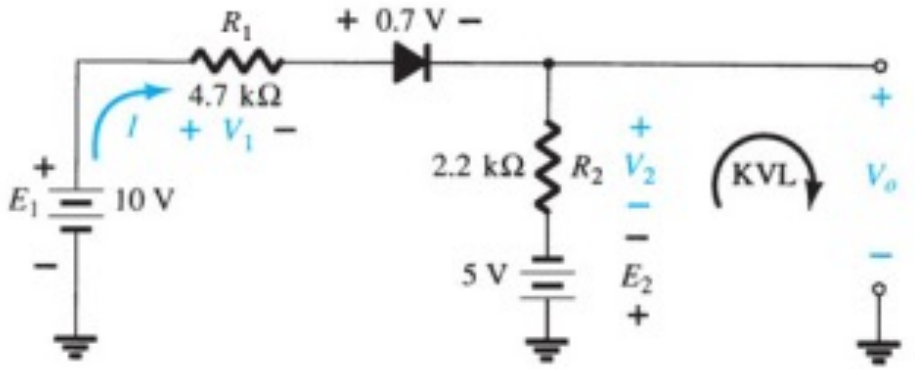
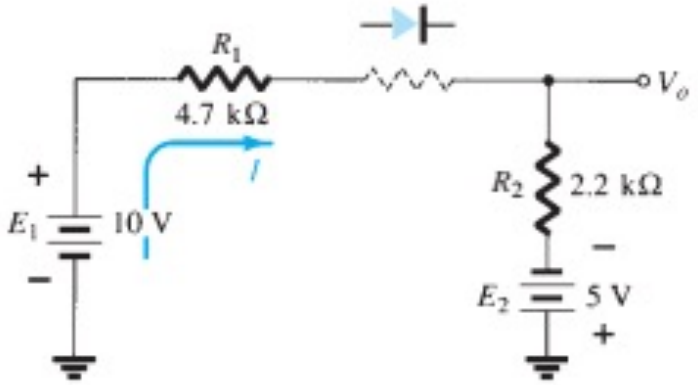


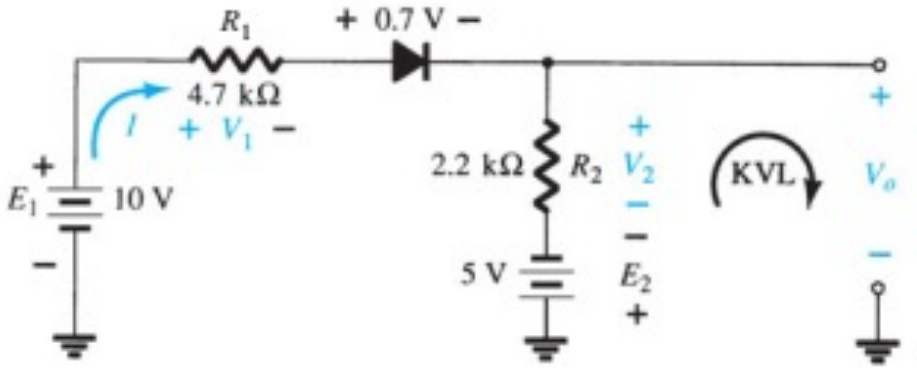
FIG. 2.25
Circuit for Example 2.9.





Series Diode Configurations

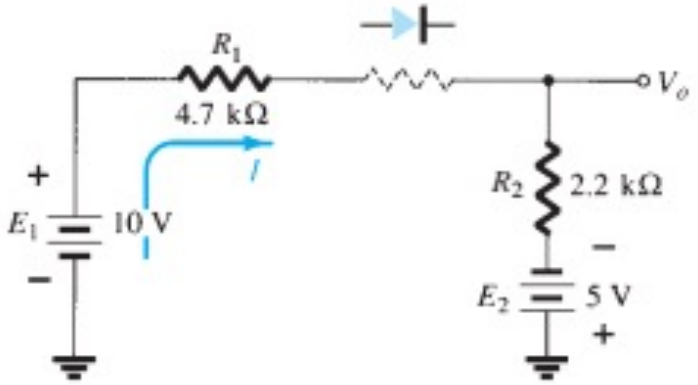


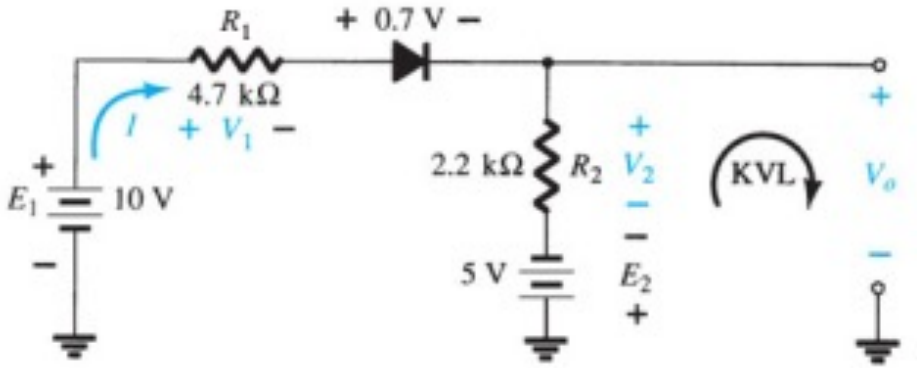


$$I = \frac{E_1 + E_2 - V_D}{R_1 + R_2} = \frac{10\,\text{V} + 5\,\text{V} - 0.7\,\text{V}}{4.7\,\text{k}\Omega + 2.2\,\text{k}\Omega} = \frac{14.3\,\text{V}}{6.9\,\text{k}\Omega}$$

$$\cong \mathbf{2.07\ mA}$$

Series Diode Configurations





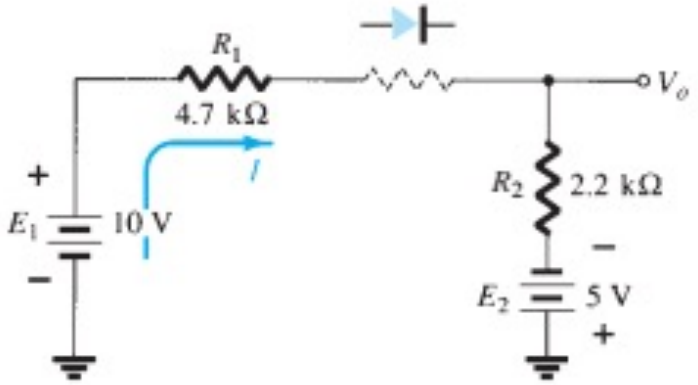
$$I = \frac{E_1 + E_2 - V_D}{R_1 + R_2} = \frac{10\,\text{V} + 5\,\text{V} - 0.7\,\text{V}}{4.7\,\text{k}\Omega + 2.2\,\text{k}\Omega} = \frac{14.3\,\text{V}}{6.9\,\text{k}\Omega}$$

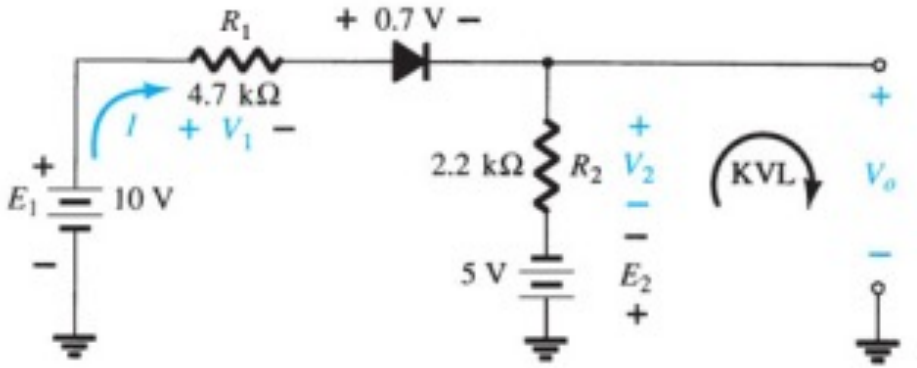
$$\cong \mathbf{2.07\ mA}$$

$$V_1 = IR_1 = (2.07\,\text{mA})(4.7\,\text{k}\Omega) = \mathbf{9.73\ V}$$

$$V_2 = IR_2 = (2.07\,\text{mA})(2.2\,\text{k}\Omega) = \mathbf{4.55\ V}$$

Series Diode Configurations





$$I = \frac{E_1 + E_2 - V_D}{R_1 + R_2} = \frac{10\,\text{V} + 5\,\text{V} - 0.7\,\text{V}}{4.7\,\text{k}\Omega + 2.2\,\text{k}\Omega} = \frac{14.3\,\text{V}}{6.9\,\text{k}\Omega}$$

$$\cong \mathbf{2.07\ mA}$$

$$V_1 = IR_1 = (2.07\,\text{mA})(4.7\,\text{k}\Omega) = \mathbf{9.73\ V}$$

$$V_2 = IR_2 = (2.07\,\text{mA})(2.2\,\text{k}\Omega) = \mathbf{4.55\ V}$$

$$-E_2 + V_2 - V_o = 0$$

$$V_o = V_2 - E_2 = 4.55\,\text{V} - 5\,\text{V} = \mathbf{-0.45\ V}$$

Parallel Configurations

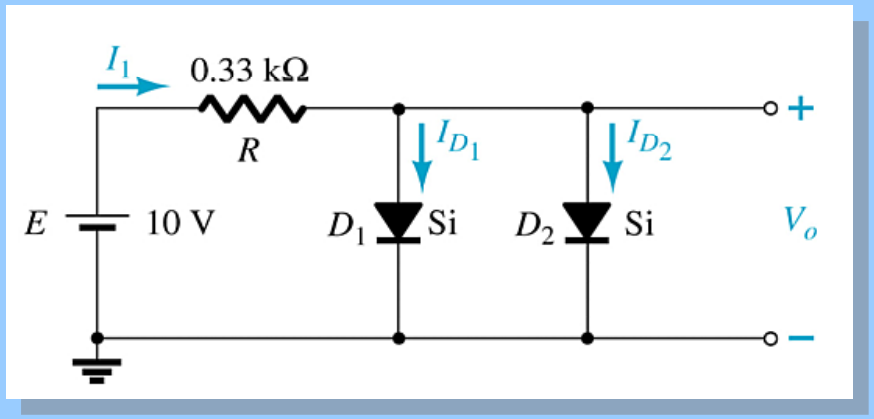
$$V_D = 0.7\,V$$

$$V_{D1} = V_{D2} = V_O = 0.7\,V$$

$$V_R = 9.3\,V$$

$$I_R = \frac{E - V_D}{R} = \frac{10\,V - .7\,V}{.33k\Omega} = 28\,mA$$

$$I_{D1} = I_{D2} = \frac{28\,mA}{2} = 14\,mA$$



Parallel Configurations

EXAMPLE 2.12 Determine the voltage V_o for the network of Fig. 2.35.

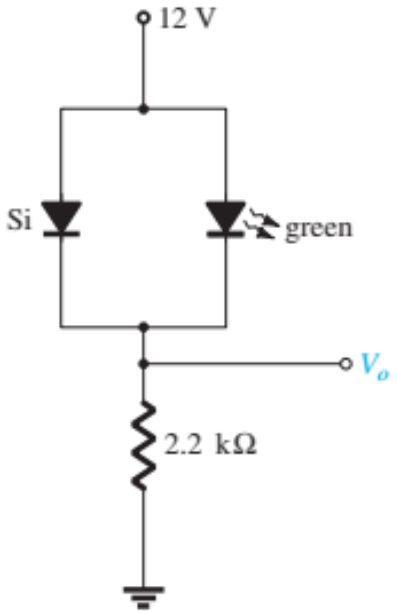


FIG. 2.35
Network for Example 2.12.

Parallel Configurations

EXAMPLE 2.12 Determine the voltage V_o for the network of Fig. 2.35.

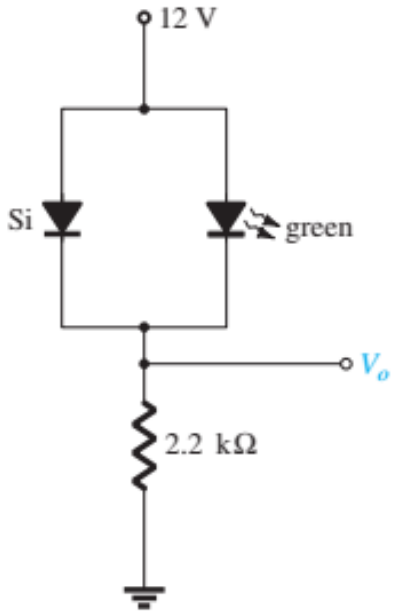


FIG. 2.35
Network for Example 2.12.

$$V_o = 12\text{ V} - 0.7\text{ V} = \mathbf{11.3\text{ V}}$$

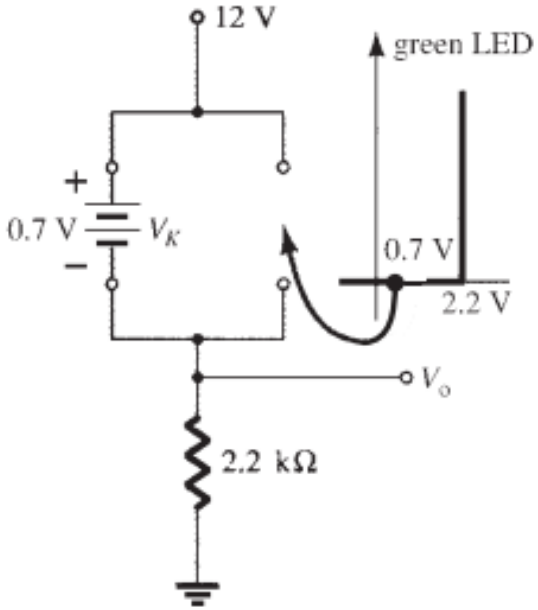


FIG. 2.36
Determining V_o for the network of Fig. 2.35.

Parallel Configurations

EXAMPLE 2.13 Determine the currents I_1, I_2, and I_{D_2} for the network of Fig. 2.37.

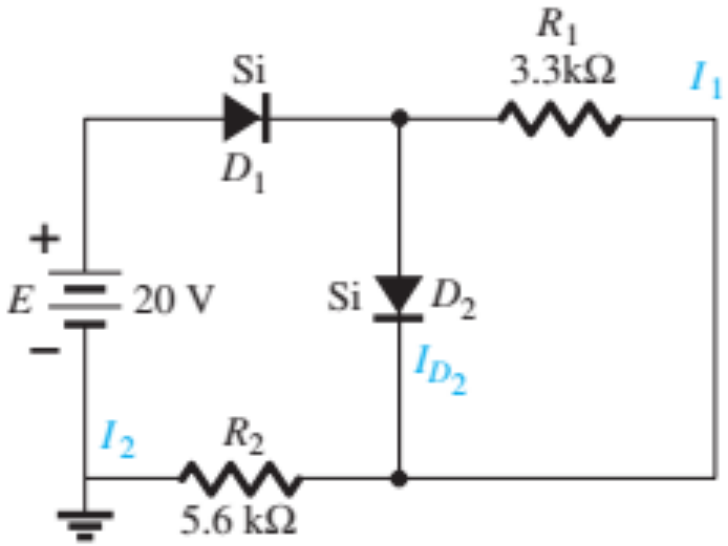


FIG. 2.37
Network for Example 2.13.

Parallel Configurations

EXAMPLE 2.13 Determine the currents I_1, I_2, and I_{D_2} for the network of Fig. 2.37.

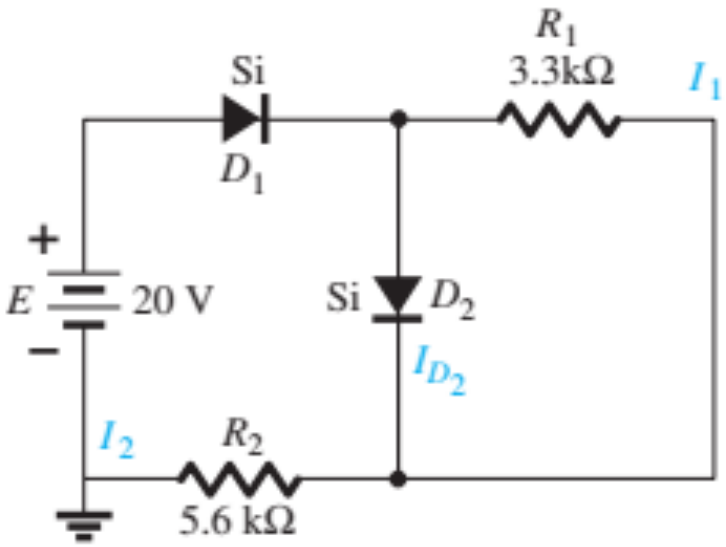


FIG. 2.37
Network for Example 2.13.

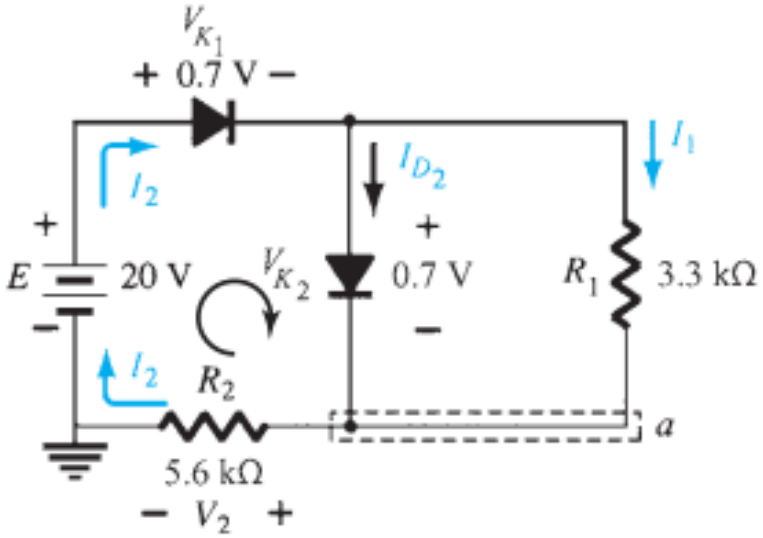


FIG. 2.38
Determining the unknown quantities for Example 2.13.

Parallel Configurations

EXAMPLE 2.13 Determine the currents I_1, I_2, and I_{D_2} for the network of Fig. 2.37.

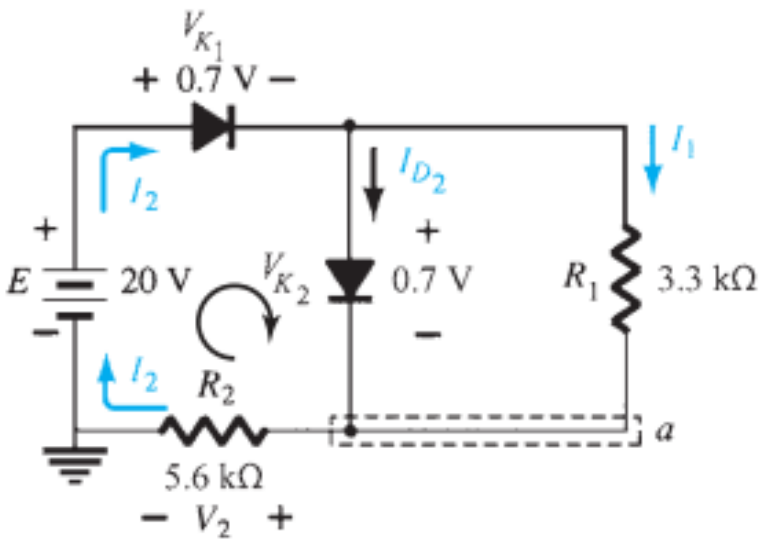


FIG. 2.38
Determining the unknown quantities for Example 2.13.

$$I_1 = \frac{V_{K_2}}{R_1} = \frac{0.7\ \text{V}}{3.3\ \text{k}\Omega} = \mathbf{0.212\ \text{mA}}$$

$$-V_2 + E - V_{K_1} - V_{K_2} = 0$$

$$V_2 = E - V_{K_1} - V_{K_2} = 20\ \text{V} - 0.7\ \text{V} - 0.7\ \text{V} = \mathbf{18.6\ \text{V}}$$

$$I_2 = \frac{V_2}{R_2} = \frac{18.6\ \text{V}}{5.6\ \text{k}\Omega} = \mathbf{3.32\ \text{mA}}$$

$$I_{D_2} + I_1 = I_2$$

$$I_{D_2} = I_2 - I_1 = 3.32\ \text{mA} - 0.212\ \text{mA} \cong \mathbf{3.11\ \text{mA}}$$